EE3408 Analogue Communications

2025

Chapter 1

Introduction

1.1 Overview

In the past, messages have been carried by runners, carrier pigeons, drumbeats, smokes and torches. These schemes were adequate for the distances and 'data rates' of the age. In most parts of the world, these modes have been superseded by electrical communication systems, with the exception of the postal service, that can transmit signals over much longer distances (even to distant planets and galaxies) and at the speed of light. A typical communication system can be modelled as shown in Figure 1.1. The components of a communication system are as follows:

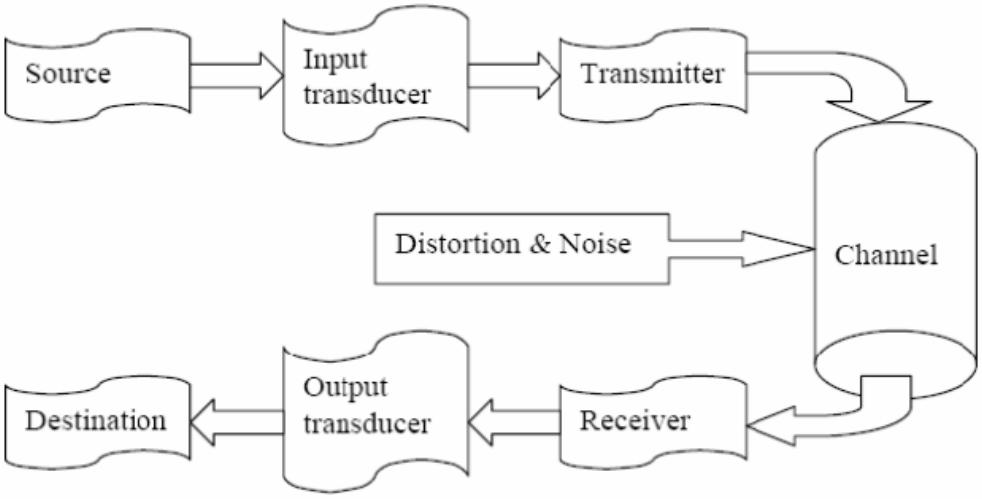


Figure 1.1: A Communication System

The *source* originates the message, such as human voice, a television picture, a teletype message, or data. If the data is non-electrical, it must be converted by an *input transducer* into an electrical waveform referred to as the *baseband signal* or the *message signal*. The information from the source will generally not be in a form that can travel through the channel, so a device called a *transmitter* modifies the baseband signal for efficient transmission. It consists of one or more of the following subsystems: a sampler, a quantizer, a coder, and a *modulator*. Similarly, the receiver may consist of a *demodulator*, a decoder, and a filter.

The *channel* is a medium - such as wire, coaxial cable, a waveguide, an optical fibre, or a radio link through which the transmitter output is sent. Sometimes a channel can carry information signal directly. For example, an audio signal can be carried directly by a twisted-pair telephone cable. On the other hand, a radio link through free space cannot be used directly for voice signals. An antenna of enormous length would be required, and it would not be possible to transmit more than one signal without interference.

The *receiver* reprocesses the signal from the channel by undoing the signal modifications made at the transmitter and the channel. The receiver output is fed to the *output transducer*, which converts the electrical signal to its original form - the message. The *destination* is the unit to which the message is communicated.

A channel acts partly as a filter to attenuate the signal and distort its waveform. The length of the channel increases its attenuation, varying from a few percent for short distances to orders of magnitude for interplanetary communication. The waveform is distorted because of different amounts of attenuation and phase shift suffered by different frequency components of the signal. For example, a square pulse is rounded or 'spread out' by the process. The signal is not only distorted by the channel, but it is also contaminated along the path by undesirable signals lumped under the broad term *noise* which are random and unpredictable from causes external and internal. External noise includes interference from signals transmitted on nearby channels; man-made noise generated by faulty contact switches for electrical equipment, by automobile ignition radiation, fluorescent lights, and natural

noise from lightning, electrical storms, solar and intergalactic radiation. With proper care, external noise can be minimized or even eliminated. Internal noise results from thermal motion of electrons in conductors, random emission, and diffusion or recombination of charged carriers in electronic devices. Proper care can reduce the effect of internal noise but can never eliminate it.

Noise is one of the basic factors that set a limit on the rate of communication. The signal-to-noise ratio (SNR) is defined as the ratio of the signal power to the noise power. The channel distorts the signal, and the noise accumulates along the path. Worse yet, the signal strength decreases while the noise level increase with distance from the transmitter. Thus, the SNR is continuously decreasing along the channel.

Amplification of the received signal to make up for the attenuation is of no avail because the noise will be amplified in the same proportion, and the SNR remains, at best, unchanged. Amplification, in practice, further deteriorates the SNR because of the added noise generated by the amplifier itself.

1.2 Need for Modulation

Baseband signals produced by various information sources are not always suitable for direct transmission over a given channel. These signals are usually further modified through a process called *modulation* to facilitate transmission. In this process, the baseband signal is used to modify some parameter of a high-frequency carrier signal. A carrier is a sinusoid of high frequency, and one of its parameters such as amplitude, frequency or phase, is varied in proportion to the baseband signal. Accordingly, we have amplitude modulation (AM), frequency modulation (FM), or phase modulation (PM). Some of the important reasons for modulation are given below:

1.2.1 Ease of radiation

For efficient transmission of electromagnetic energy, the radiating antenna should be of the order of one-tenth or more of the wavelength of the signal radiated. For many baseband signals, the wavelengths are too large for practical antenna dimensions.

Example: The power in a speech signal is concentrated at frequencies in the range 100 Hz to 3000 Hz. The corresponding wavelength is 3000 km to 100 km, necessitating an impractically large antenna. Instead, if we modulate a 100 MHz carrier (with a wavelength of only 3m), we require an antenna whose size is of the order of 30cm.

1.2.2 Simultaneous transmission of several signals

We can use various signals to modulate different carrier frequencies, thus translating each signal to a different frequency range. If the carriers are chosen sufficiently far apart in frequency, the spectra of the modulated signals will not overlap and thus will not interfere with each other. This method of transmitting several signals simultaneously is known as frequency-division multiplexing.

1.3 Analog and Digital Messages

Messages are either digital or analog; digital messages are constructed with a finite number of symbols. For example, printed language (text) consists of 26 letters, 10 numbers, a space, and several punctuation marks. Thus a text is a digital message constructed from about 50 symbols. Similarly, a Morse-coded telegraph message is a digital message constructed from a set of only two symbols: mark and space. It is therefore a binary message, implying only two symbols. A digital message constructed with M symbols is called an M-ary message.

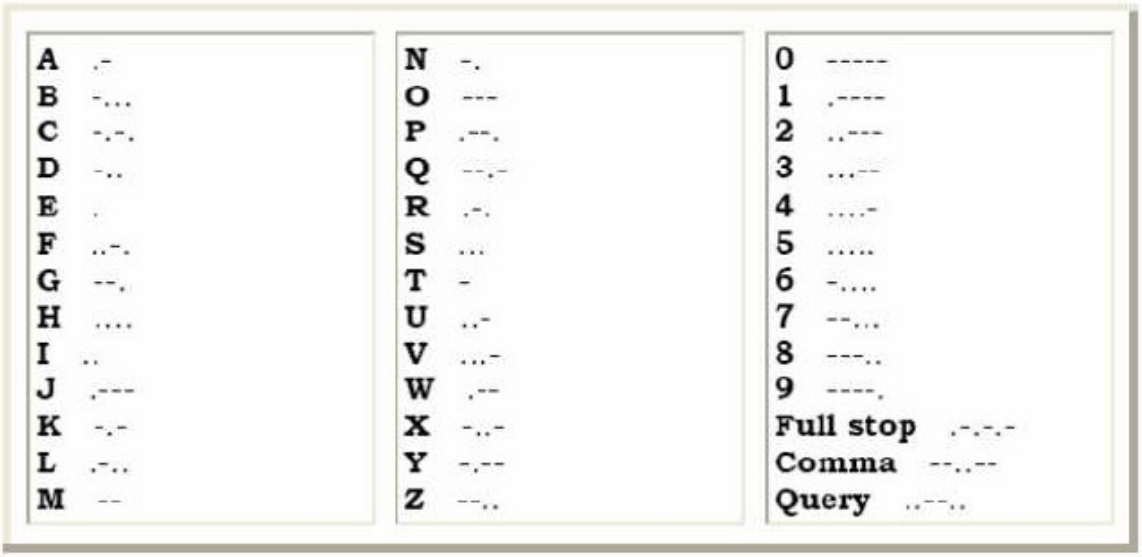


Figure 1.2: Morse Code

On the other hand, analog messages are characterized by data whose value varies over a continuous range. For example, the temperature or the atmospheric pressure of a certain location can vary over a continuous range and can assume infinite possible values. Similarly, a speech waveform has amplitudes that vary over a continuous range. Over a given time interval, infinite possible different speech waveforms exist, in contrast to only a finite number of possible digital messages. Digital messages are transmitted by using a finite set of electrical waveforms. For instance, in the Morse code, an electrical pulse of amplitude A/2 can transmit a mark, and a pulse of amplitude -A/2 can transmit a space. In an M-ary case, M distinct electrical pulses (or waveforms) are used; each of the M pulses represents one of the M possible symbols.

The task at the receiver is to extract a message from a distorted and noisy signal at the channel output. Message extraction is often easier from digital signals than from analog signals. Consider a binary case as shown in Figure 1.3. Two symbols are encoded as rectangular pulses of amplitudes A/2 and -A/2. The only decision at the receiver for the distorted data is the selection between two possible pulses received, not the details of the pulse shape; the decision is readily made with reasonable certainty even if the pulses are distorted and noisy. Hence a digital communication system can transmit messages with greater accuracy than analogue system in the presence of distortion and noise.

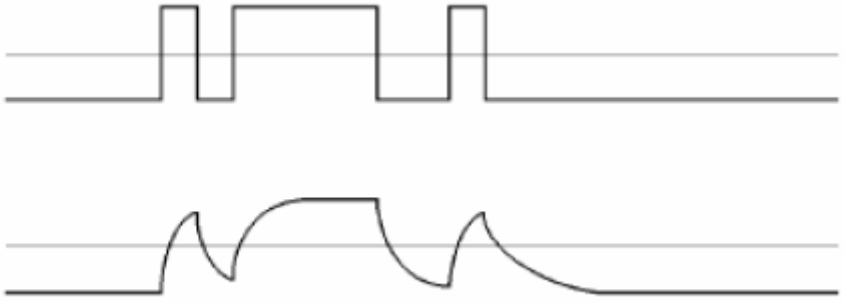


Figure 1.3: Distorted Digital Data

The possibility of using regenerative repeaters is a further advantage for digital communication. A repeater station detects pulses and transmits new clean pulses, thus combating the accumulation of distortion and noise and enabling information transmission over longer distances with greater accuracy.

In contrast to digital messages, the waveform in analog messages is important, and even a slight distortion or interference in the waveform will cause an error in the received signal. A further difficulty is that a regenerative repeater is not viable for analog signals because the noise and distortion, no matter how small, cannot be cleaned up from the signal. As a result, the distance over which an analog message can be transmitted is limited by the transmitter power. Yet analog communication is being used widely and successfully despite these problems. A tendency exists, however, to replace analog

systems with digital systems as the latter become more economical because of dramatic cost reduction achieved in the fabrication of digital circuitry.

1.4 Decibels

Decibels are used extensively in all electronic communications to represent relative parameters such as gain or loss and absolute parameters such as signal amplitude in volts and watts. In general, the number decibels (dB) can be determined from the following general relationship:

$$G(dB) = \ 10\ log\frac{P_1}{P_2} \tag{1.1}$$

where P_1 and P_2 are power levels in Watts. Substituting $P_1 = (V_1)^2/R_1$ and $P_2 = (V_2)^2/R_2$ into the above expression gives

$$G(dB) = \ 20\ log_{10}\frac{V_1}{V_2} + \ 10\ log_{10}\frac{R_1}{R_2} \tag{1.2}$$

where V_1 and V_2 are voltage amplitudes in Volts and R_1 and R_2 are the respective resistances in Ohms. When $R_1 = R_2$, the second term is zero and the above expression reduces to

$$G(dB) = \ 20\ log_{10}\frac{V_1}{V_2} \tag{1.3}$$

Decibels are relative because any values for P_1 and P_2 will give the same results as long as their ratio is the same. Decibels are generally used to represent the gain or loss in amplifiers and other electronic circuits. The gain in decibels of an amplifier can be determined by the above relationships. Decibels are also used to represent the loss between transmission media. Positive decibel values indicate a gain, and negative values indicate a loss.

Any signal level in watts can be converted into decibels, using 1W as the reference value. Since theresulting value is referenced to 1W, then dBW is used as the units for such a system. In general,

$$P(dBW) = \ 10\ log_{10}\frac{P}{1W} \tag{1.4}$$

Decibels-watts (dBW) are used extensively in communications to represent signal amplitudes in a system. For example, the power radiated by a typical satellite antenna may be 1000 W, or +30dBW, whereas the power received at the earth station is approximately $1x10^{-17}\ W$, or -170 dBW. Negative values of dBW occur when the signal amplitude is less that the reference value of 1W.

The dBm system uses 1mW as a reference

$$P(dBm) = \ 10\ log_{10}\frac{P}{1mW} = \ 10\ log_{10}\frac{P}{1x10^{-3}\ W} \tag{1.5}$$

Conversion between dBW and dBm is shown below:

$$dBm = dBW + 30dB \tag{1.6}$$

1.5 Noise

Noise in communication systems originates both in the channel and in the communication equipment. Noise consists of undesired, usually random, variations that interfere with the desired signals and inhibit communication. It cannot be avoided completely, but its effects can be reduced by various means, such as reducing the signal bandwidth, increasing the transmitter power, and using low-noise amplifiers for weak signals. It is helpful to divide noise into two types: internal noise, which originates within the communication equipment, and external noise, which is a property of the channel.

1.5.1 External Noise

If the channel is a radio link, there are many possible sources of noise. The reader is no doubt familiar with the 'static' that afflicts AM radio broadcasting during thunderstorms. Interference from automobile ignition systems is another common problem. Domestic and industrial electrical equipment, from light dimmers to vacuum cleaners to computers, can also cause objectionable noise. Not so obvious, but of equal concern to the communications specialist, is the noise produced by the sun and the stars.

Equipment noise - Noise is generated by equipment that produces sparks. Noise of this type has a broad frequency spectrum, but its energy is not equally distributed over the frequency range. This type of interference is generally more severe at lower frequencies. Man-made noise can propagate through space or along power lines. A typical solution for a computer, for instance, involves shielding and grounding the case and all connecting cables and installing a low-pass filter on the power line where it enters the enclosure.

Atmospheric noise - Atmospheric noise is often called static because lightning, which is a static-electricity discharge, is a principal source of atmospheric noise. This type of disturbance can propagate for long distances through space. Most of the energy of lightning is found at relatively low frequencies, up to several megahertz.

Obviously nothing can be done to reduce atmospheric noise at the source. However, circuits are available to reduce its effect by taking advantage of the fact that this noise has a very high peak-to-average power ratio; that is, the noise occurs in short intense bursts with relatively long periods of time between bursts. It is often possible to improve communication by simply disabling the receiver for the duration of the burst. This technique is called noise blanking.

Space noise - The sun is a powerful source of radiation over a wide range of frequencies, including radio-frequency spectrum. Other stars also radiate noise called cosmic, stellar, or sky noise. Its intensity when received on the earth is naturally much less than for solar noise because of the greater distance.

Solar noise can be a serious problem for satellite reception, which becomes impossible when the satellite is in a line between the antenna and the sun. Space noise is more important at higher frequencies (VHF and above) because atmospheric noise dominates at lower frequencies and because most of the space noise at lower frequencies is absorbed by upper atmosphere.

1.5.2 Internal Noise

Noise is generated in all electronic equipment. Both passive (such as resistors and cables) and active devices (such as diodes, transistors, and tubes) can be noise sources. Several types of noise will be examined in this section, beginning with the most important, thermal noise.

Thermal noise - Thermal noise is produced by the random motion of electrons in a conductor due to heat. The term 'noise' is often used alone to refer to this type of noise, which is found everywhere in electronic circuitry. The power density of thermal noise is constant with frequency, from zero to frequencies well above those used in electronic circuits. That is, there is equal power in every hertz of bandwidth. Thermal noise is thus an equal mixture of noise of all frequencies. It is sometimes called *white noise*, in analogy with white light, which is an equal mixture of all colours.

The power available from a conductor is a function of its temperature, as shown by the equation:

$$P_N = kTB \tag{1.7}$$

where P_N = noise power in watts; k = Boltzmann's constant, $1.38x10^{-23}$ (J/K); T = absolute temperature in kelvins (K), this can be found by adding 273 to the celcius temperature; B = noise power bandwidth in hertz (Hz).

Equation 1.7 shows that the noise power is directly related to bandwidth over which the noise is observed (the noise power bandwidth). For instance, a resistor connected across the terminals of a receiver (with a noise power bandwidth of 10 kHz) at a temperature of 27^0 *C* will contribute a noise power of $P_N = 4.14x10^{-17}$ *W*. This is not a great deal of power, but it can be significant at the signal levels dealt with in sensitive receiving equipment.

Thermal noise power exists in all conductors and resistors at any temperature above absolute zero. The only way to reduce it is to decrease the temperature or the bandwidth of a circuit (or both). At any instant in time, more electrons are moving in some directions than in others. The movement of each electron constitutes a very small current and hence the net current fluctuates continuously in magnitude and direction. This net current flows in the resistance of the conductor and produces a *noise voltage*. Over a period of time the average noise voltage is zero but the root mean square (rms) noise voltage is developed, given by:

$$V_N = \sqrt{4kTBR} \tag{1.8}$$

Shot noise - This type of noise has a power spectrum that resembles that of thermal noise, having equal energy in every hertz of bandwidth, at frequencies from dc into the gigahertz region. However, the mechanism that creates shot noise is different. Shot noise is due to random variations in current flow in active devices such as transistors and semiconductor diodes.

Shot noise is caused by the random movement of electrons and holes across each p-n junction. The rms value value of the shot noise current is:

$$I_N = \sqrt{2qI_0B} \tag{1.9}$$

where I_N = rms noise current (in amperes); q = magnitude of the electronic charge ($1.6x10^{-19}$ C); I_0 = dc current flowing across the junction, in amperes; and B = bandwidth (Hz) over which the noise is observed.

Note: All of the noises we have looked at have random waveforms. The amplitude at a particular time cannot be predicted, though the average voltage is zero and the rms voltage or current and average power are predictable (and *not zero*). Signals like these are said to be uncorrelated; that is, they have no definable phase relationship. In such a case, voltages and currents do not add directly, but the total voltage (of series circuits) or current (of parallel circuits) can be found by taking the square root of the sum of squares of the individual voltages or currents. Mathematically, we can say, for voltage sources in series:

$$V_{N_t} = \sqrt{V_{N_1}^2 + V_{N_2}^2 + V_{N_3}^2 + \ldots} \tag{1.10}$$

and similarly, for current sources in parallel:

$$I_{N_t} = \sqrt{I_{N_1}^2 + I_{N_2}^2 + I_{N_3}^2 + \ldots} \tag{1.11}$$

Partition noise – It is similar to shot noise in its spectrum and mechanism of generation, but it occurs only in devices where single current separates into two or more paths. An example is a bipolar junction transistor, where the emitter current is the sum of the collector and base currents. As the charge carriers divide into one stream of the other, a random element in the currents is produced.

Excess noise – It is also called *Flicker noise* or *1/f noise* because the noise power varies inversely with frequency. Sometimes it is called *Pink noise* because there is proportionally more energy at the low-frequency end of the spectrum than with white noise, just as pink light has a high proportion of red (the low-frequency end of the visible spectrum) than white light.

Excess noise is found in tubes, but is a more serious problem in semiconductors and in carbon resistors. Excess noise is rarely a problem in communication circuits because it declines with increasing frequency and is usually insignificant above approximately 1 kHz.

1.6 Filters

In electronic communications systems, it is often necessary to separate a specific range of frequencies from the total spectrum. This is normally achieved with filters. A *filter* is an electrical network that alters the amplitude and/or phase characteristics of a signal with respect to frequency. There are four basic types of filters, low-pass, high-pass, band-pass and band-stop. A *low-pass* filter is designed to pass all frequencies below the cut-off frequency and reject all frequencies above the cut-off frequency. A *high-pass* filter is designed to pass all frequencies above the cut-off frequency and reject all frequencies below the cut-off frequency. A *band-pass* filter passes all frequencies within a band of frequencies and rejects all other frequencies outside the band. A *band-stop* filter rejects all frequencies within a band of frequencies and passes all other frequencies outside the band. A band-stop filter is often referred to as a *notch filter*.

Since filters are defined by their frequency-domain effects on signals, it makes sense that the most useful analytical and graphical descriptions of filters also fall into the frequency domain. The most common way to describe the frequency response characteristics of a filter is to plot the filter voltage gain in dB as a function of frequency (f).

$$A(dB) = 10\log\frac{P_{out}}{P_{in}} = 20\log\frac{V_{out}}{V_{in}} \tag{1.12}$$

The frequency at which the output power gain drops to 50% of the maximum value is called the *cut-off frequency*. When the output power gain drops to 50%, the voltage gain drops 3 dB (0.707 of the maximum value). When the filter dB voltage gain is plotted as a function of frequency on a semilog graph using straight line to approximate the actual frequency response, it is called a *Bode plot*. A Bode plot is an ideal plot of filter frequency response because it assumes that the voltage gain remains constant in the passband until the cut-off frequency is reached, and then drops in a straight line.

The frequency-domain behavior of a filter is also described mathematically in terms of its *transfer function*. This is the ratio of the Laplace transforms of its output and input signals. The voltage transfer function $H(s)$ of a filter can therefore be written as:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} \tag{1.13}$$

The *order* of a filter is the highest power of the variable s in its transfer function. The order of a filter is usually equal to the total number of capacitors and inductors in the circuit. (A capacitor built by combining two or more individual capacitors is still one capacitor.) Higher-order filters will obviously be more expensive to build, since they use more components, and they will also be more complicated to design. However, higher-order filters can more effectively discriminate between signals at different frequencies.

The range of frequencies passed by a filter is known as the filter's *passband*. The *stopband* is the range of frequencies over which unwanted signals are attenuated. The rate of change of attenuation (known as *roll-off*) between the passband and the stopband also differs from one filter to the next. The slope of the curve in this region depends strongly on the order of the filter, with higher-order filters having steeper cutoff slopes. The attenuation slope is usually expressed in dB/octave (an octave is a factor of 2 in frequency) or dB/decade (a decade is a factor of 10 in frequency).

1.6.1 Passive Filters

Filters made up of passive components: resistors, capacitors, and inductors, are referred to as *passive filters*. A passive filter is simply a filter that uses no amplifying elements (transistors, operational amplifiers, etc). Because they have no active components, passive filters require no power supplies. Since they are not restricted by the bandwidth limitations of op amps, they can work well at very high frequencies. They can be used in applications involving larger current or voltage levels than can be handled by active devices. Passive filters also generate little noise when compared with circuits using active gain elements. The noise that they produce is simply the thermal noise from the resistive components, and, with careful design, the amplitude of this noise can be very low.

Passive filters have some important disadvantages in certain applications, however. Since they use no active elements, they cannot provide signal gain. Inductors are necessary for the synthesis of most useful passive filter characteristics, and these can be prohibitively expensive if high accuracy (1% or 2%, for example), small physical size, or large values are required. Standard values of inductors are not very closely spaced, and it is difficult to find an off-the-shelf unit within 10% of any arbitrary value, so adjustable inductors are often used. Tuning these to the required values is time-consuming and expensive when producing large quantities of filters. Furthermore, complex passive filters (higher than 2nd-order) can be difficult and time-consuming to design.

Bandpass Filters - As an example of a 2nd order bandpass, the network of Figure 1.4 has the transfer function:

$$H(s) = \frac{s}{s^2 + s + 1} \tag{1.14}$$

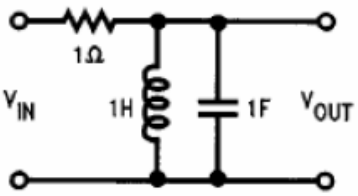


Figure 1.4: Example of a Bandpass filter network

Several examples of bandpass amplitude response curves are shown in Figure 1.5. Note that while some bandpass responses are very smooth, other have *ripple* (gain variations in their passbands. Other have ripple in their stopbands as well. Bandpass filters have two stopbands, one above and one below the passband.

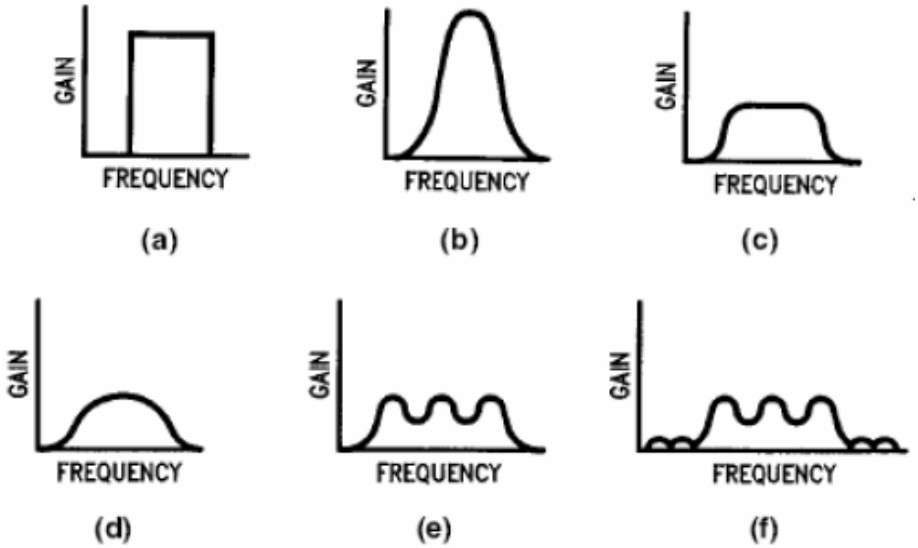


Figure 1.5: Examples of Bandpass filter Amplitude Responses

The precise shape of a band-pass filter's amplitude response curve will depend on the particular network, but any band-pass response will have a peak value at the filter's *center frequency*. The center frequency is equal to the geometric mean of the lower and the upper -3 dB frequencies:

$$f_c = \sqrt{f_l f_h} \tag{1.15}$$

Another quantity used to describe the performance of a filter is the filter's "Q". This is a measure of the "sharpness" of the amplitude response. The Q of a band-pass filter is the ratio of the center frequency to the difference between the -3 dB frequencies (also known as the *-3 dB bandwidth*). Therefore:

$$Q = \frac{f_c}{f_h - f_l} \tag{1.16}$$

Bandpass filters are used in electronic systems to separate a signal at one frequency or within a band of frequencies from signals at other frequencies. Such a filter could also reject unwanted signals at other frequencies outside of the passband, so it could be useful in situations where the signal of interest has been contaminated by signals at a number of different frequencies.

Notch Filters - A filter with effectively the opposite function of the bandpass is the *bandstop* or *notch filter*. As an example, the components in the network of Figure 1.6 form the notch filter, which has the transfer function:

$$H(s) = \frac{s^2 + 1}{s^2 + s + 1} \tag{1.17}$$

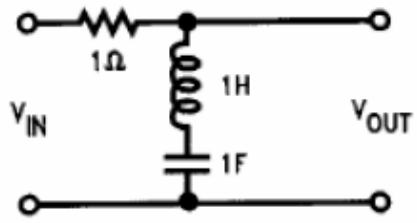


Figure 1.6: Example of a Notch filter

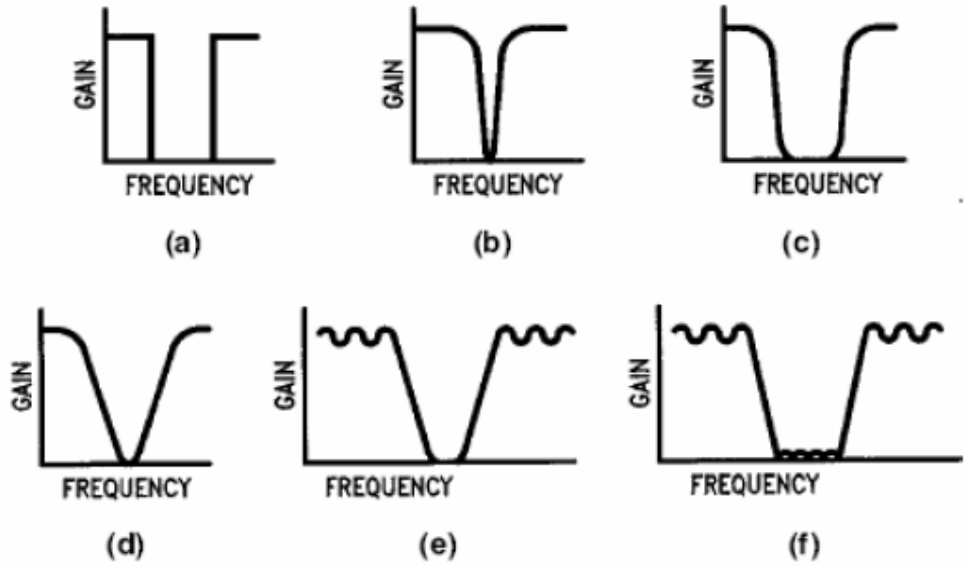


Figure 1.7: Examples of Notch filter Amplitude Responses

Notch filters are used to remove an unwanted frequency from a signal, while affecting all other frequencies as little as possible. An example of the use of a notch filter is with an audio program that has been contaminated by 60 Hz powerline hum. A notch filter with a center frequency of 60 Hz can remove the hum while having little effect on the audio signals. A number of notch filter amplitude response curves are shown in Figure 1.7 where curve (a) shows an ideal" notch response, while the other curves show various approximations to the ideal characteristic.

Lowpass Filters - Figure 1.8 shows an example of a 2nd order lowpass filter, with the following transfer function:

$$H(s) = \frac{1}{s^2 + s + 1} \tag{1.18}$$

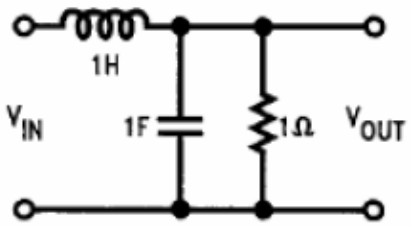


Figure 1.8: Example of a Lowpass filter

If we let $s = j\omega$, it is easy to see by inspection that this transfer function has more gain at low frequencies than at high frequencies. As ω approaches 0, H approaches 1; as ω approaches infinity, H approaches 0. Note that the various approximations to the unrealizable ideal low-pass amplitude characteristics take different forms, some being monotonic (always having a negative slope), and others having ripple in the passband and/or stopband.

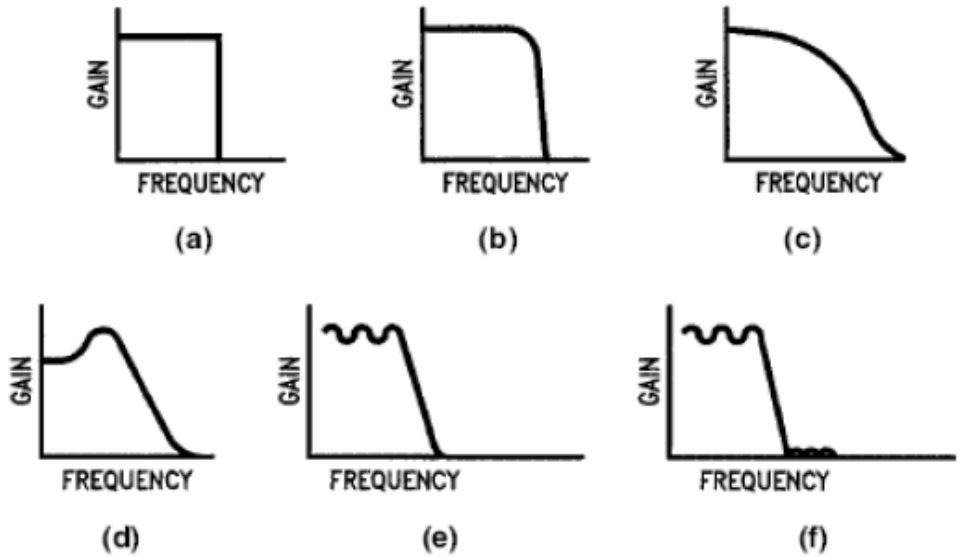


Figure 1.9: Examples of Lowpass filter Amplitude Response Curves

Low-pass filters are used whenever high frequency components must be removed from a signal.

Highpass Filters - An example of a 2nd order highpass filter with following the transfer function is shown in Figure 1.10.

$$H(s) = \frac{s^2}{s^2 + s + 1} \tag{1.19}$$

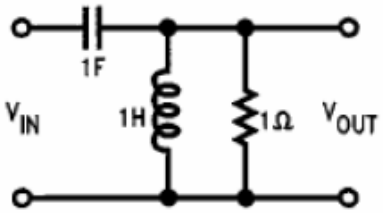


Figure 1.10: Example of a Highpass filter

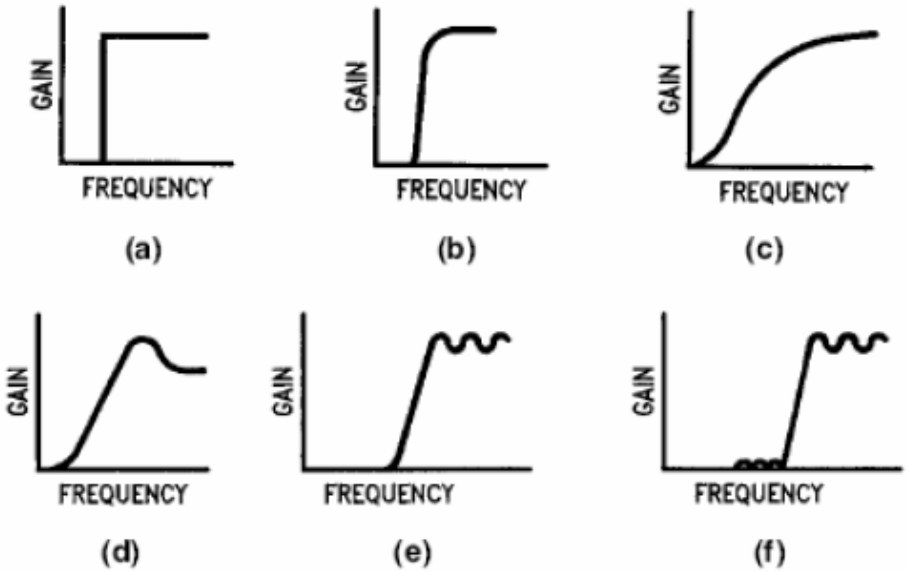


Figure 1.11: Examples of Highpass filter Amplitude Response Curves

Let's take another look at the transfer function equations and response curves presented so far. First note that all of the transfer functions share the same denominator. Also note that all of the numerators are made up of terms found in the denominator: the high-pass numerator is the first term (s^2) in the denominator, the bandpass numerator is the second term (s), the low-pass numerator is the third term (1), and the notch numerator is the sum of the denominator's first and third terms ($s^2 + 1$).

Second-order filters are characterized by four basic properties: the filter *type* (high-pass, bandpass, etc.), the *passband gain* (all the filters discussed so far have unity gain in the passband, but in general filters can be built with any gain), the *center frequency* (f_0 and the *filter Q*. Q was mentioned earlier in connection with bandpass filters, but in second- order filters it is also a useful quantity for describing the behavior of the other types as well. The Q of a second-order filter of a given type will determine the relative shape of the amplitude response. Q can be found from the denominator of the transfer function if the denominator is written in the form:

$$D(s) = s^2 + \frac{\omega_0}{Q}s + \omega_0^2 \tag{1.20}$$

As was noted in the case of the bandpass and notch functions, Q relates to the "sharpness" of the amplitude response curve. As Q increases, so does the sharpness of the response. Low-pass and high-pass filters exhibit "peaks" in their response curves when Q becomes large.

1.6.2 Active Filters

Active filters use amplifying elements, especially op amps, with resistors and capacitors in their feedback loops, to synthesize the desired filter characteristics. Active filters can have virtually any arbitrary gain. They are also usually easier to design than passive filters. Possibly their most important attribute is that they lack inductors, thereby reducing the problems associated with those components. Still, the problems of accuracy and value spacing also affect capacitors, although to a lesser degree. Performance at high frequencies is limited by the gain-bandwidth product of the amplifying elements, but within the amplifier's operating frequency range, the op amp-based active filter can achieve very good accuracy, provided that low-tolerance resistors and capacitors are used. Active filters will generate noise due to the amplifying circuitry, but this can be minimized by the use of low-noise amplifiers and careful circuit design.

1.6.3 Crystal and Ceramic Filters

A crystal (usually quartz) or ceramic material is *piezo-electric*. This kind of material is a natural transducer if you connect a set of terminals to a small block of the crystal or ceramic material, you get a voltage when you squeeze or twist the block; alternatively, the block twists or changes shape if you put an external voltage across it. So it naturally changes mechanical movement into an electrical signal, or vice versa.

The properties of a quartz crystal, namely very high Q and excellent stability, can also be used to advantage in the design of bandpass filters. Using only one crystal would make the bandwidth too narrow for most purposes, but several crystals, tuned to slightly different frequencies, can be used in a crystal lattice to make a bandpass filter with a bandwidth of several kilohertz and an excellent shape factor.

Some ceramics exhibit a piezoelectric effect similar to that of quartz and can be used as filters as well. The Q of ceramic filters is lower than for quartz, and the bandwidth is wider. They are quite useful for wideband signals, such as broadcast FM. Crystal and ceramic filters have the additional advantage, compared with conventional tuned circuits, or requiring no adjustment.

Chapter 2

Signal Representations

2.1 Introduction

In the context of communications, a signal is basically some information somehow encoded as a wave. Everything travels as a wave, so let's look at some of the simple ideas of that. Figure 2.1 presents a basic sine wave. The vertical axis is the volume, or air-pressure, or strength, or energy, etc., depending on what this wave is. For example, if this is an ocean wave, then the vertical axis may be the wave height. If this is a sound wave in the air, then it is the air-pressure. If it is the sound wave in a wire, then it's the voltage. The horizontal axis (here depicted in terms of π) often represents the time or distance (or space) that the wave traverses in. Waves generally have a *period*, which can be viewed as the time/distance of when the wave repeats itself/starts the next iteration. In Figure 2.1 the period is 2π. A closely related property of the period is the frequency. *Frequency* is the speed at which the wave cycles its periods, and is always expressed in proper speed units, as in `cycles per second', commonly known as Hertz (Hz), and `kilo-cycles per second' (commonly known as kHz).

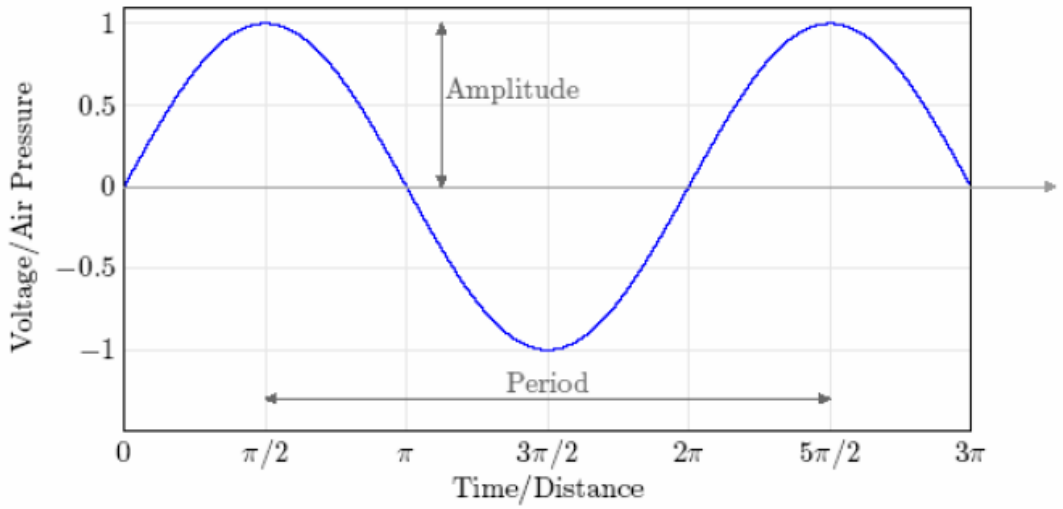


Figure 2.1: An Analog Sinewave

A thorough understanding of the techniques used to represent and display electrical signals is needed to analyze and design communication systems. The signal-analysis techniques presented in this chapter are especially important to understanding the various modulation concepts that are described in the next chapter.

A sine wave, which is the simplest type of electrical signal, is usually represented by the following mathematical relationship.

$$v = V\sin(\omega t + \phi) \tag{2.1}$$

where v = voltage at any time t, V = maximum voltage (amplitude) in volts, $\omega = 2\pi f$ in radians per second, f = frequency in Hertz and ϕ =phase angle in radians.

Sine (or cosine) waves are called *sinusoidal* signals. Sinusoidal waves are very important because any non-sinusoidal signal can be represented by a series of sinusoidal waves. Non-sinusoidal signals are called *complex waves*. Square waves and sawtooth waves are examples of complex signals. Audio and video waves are also examples of complex signals.

Signals that repeat after an interval of time are called ***periodic*** signals (waves). Thus, the signal $f(t)$ is periodic if there is a number T such that $f(t + T) = f(t)$ for all t. Periodic waves are also called *repetitive*

waves. The smallest positive number T (time interval) after which the signal repeats itself exactly is called the PERIOD. The period defines the duration of one complete cycle of $f(t)$. The sine, square, and sawtooth waves are all periodic because they repeat after an interval of time.

Any signal for which there is no value of T after which the signal repeats itself exactly is said to be ***non-periodic*** or ***aperiodic***. Audio and video signals are not periodic because they change continuously.

Energy Signals, Power Signals

An energy signal is a pulse-like signal that usually exists for only a finite interval of time or even if present for an infinite amount of time, at least has a major portion of its energy concentrated in a finite time interval.

An energy signal is defined to be the one for which equation

$$E = \int_{t_1}^{t_2} |f(t)|^2 \, dt \tag{2.2}$$

is finite even when the time interval becomes infinite. i.e. when

$$\int_{-\infty}^{\infty} |f(t)|^2 \, dt < \infty \tag{2.3}$$

Examples of energy signals are illustrated in Figure 1.1

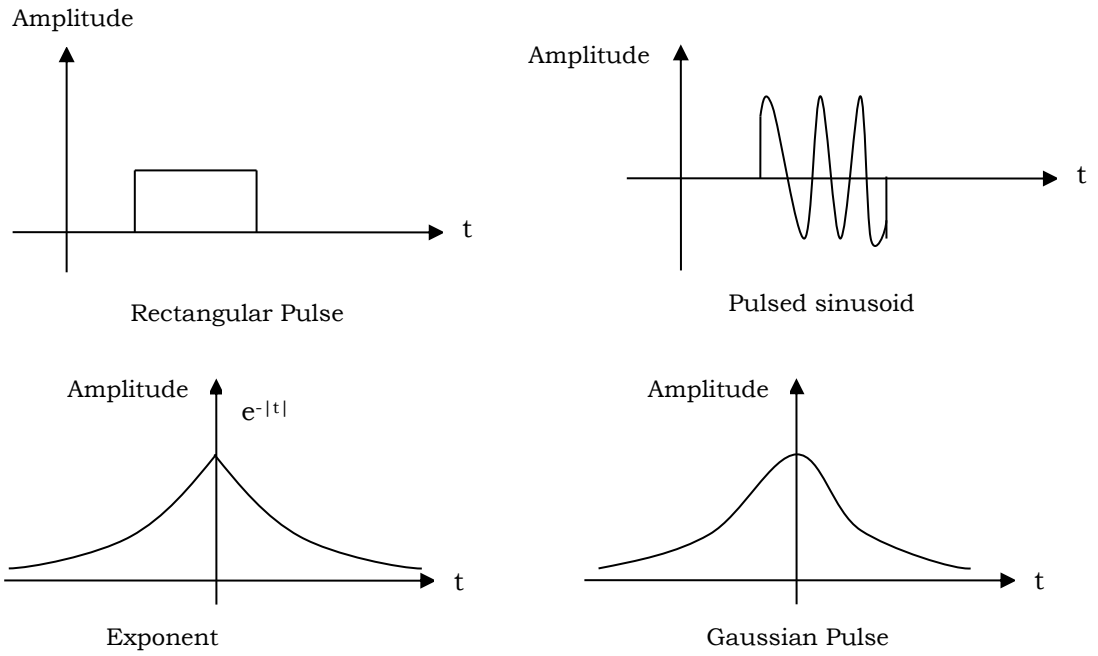


Figure 1.1: Some Energy Signals

The power signal is the signal *f(t)* which has a finite average power even when the time interval becomes infinite.

$$P = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |f(t)|^2 \, dt \tag{2.4}$$

The RHS remains finite, but non-zero when the interval becomes infinite. i.e.

$$0 \leq \lim_{T \to \infty} \frac{1}{T} \int_{t_1}^{t_2} |f(t)|^2 \, dt < \infty \tag{2.5}$$

2.2 Time and Frequency Domains

Time domain is simply another name for displaying or defining the amplitude of some electrical quantity such as voltage, current or power versus time. An oscilloscope provides a time domain display of the input signal applied to its input terminals. *Frequency domain* refers to the displaying of some electrical quantity versus frequency. A display in the frequency domain normally is called a *spectrum*. Figure 2.2 illustrates a sine wave at 4Hz. Notice that the horizontal axis is now time, and is expressed in seconds. You can count the number of times the wave cycles, and you'll see that it's 4, i.e. 4 Hz.

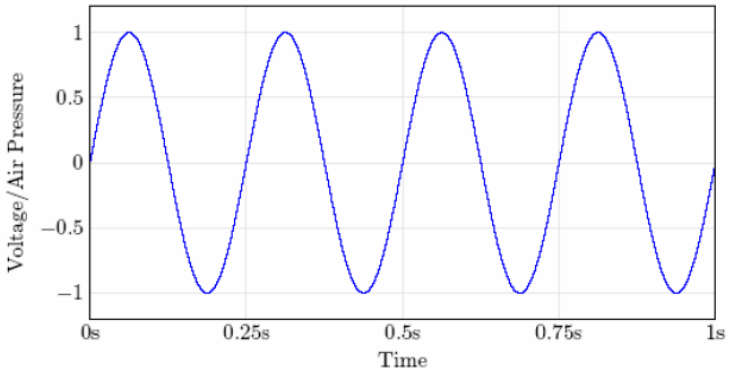


Figure 2.2: A 4 Hz sine wave

The radian frequency (ω) can be found from the time-domain representation of the signal by finding the period (that is, the time T after which the whole signal repeats exactly) and using the equations:

$$f = \frac{1}{T} \tag{2.6}$$

and

$$\omega = 2\pi f \tag{2.7}$$

Similarly, knowing the frequency, you can find the period. Another closely related value is the *wavelength*, which is basically the distance occupied by the wave period. Given the speed of propagation (light in a vacuum, or sound in the air, etc.), we can calculate that too. For example, light/electricity travel around $3x10^8$ meters per second. Let's imagine that Figure 2.2 represents electricity in a wire, then the wavelength is just $3x10^8 \div 4$ or just $7.5x10^7$ meters, or simply:

$$\lambda = \frac{velocity}{frequency} \tag{2.8}$$

Obviously you can find velocity if you know the wavelength and frequency. Continuing on with the discussion, Figure 2.3 shows a wave of 12Hz, and with $\frac{1}{3}$ amplitude. Notice how the period got smaller as the frequency increased.

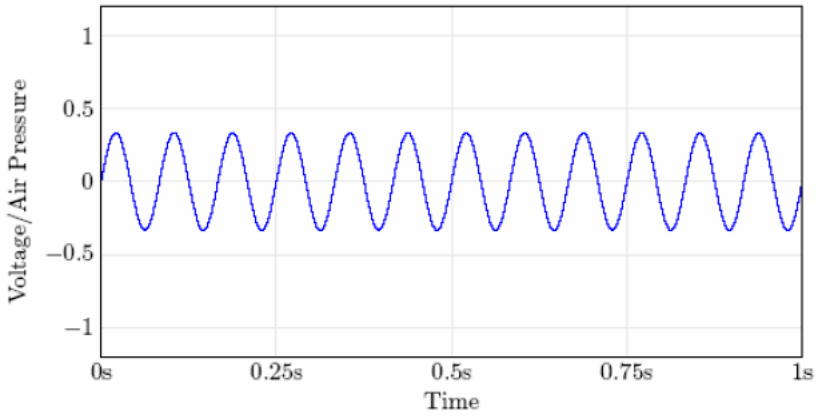


Figure 2.3: A 12 Hz sine wave

Sinusoidal signals are used as the basis for all electrical signals because any non-sinusoidal signal can be represented by the sum of a series of sine and cosine signals. Figure 2.4 illustrates the idea of taking two waves and adding them together. In this case, we've added waves from Figures 2.2 and 2.3. The frequency domain of Figure 2.4 might look something like Figure 2.5.

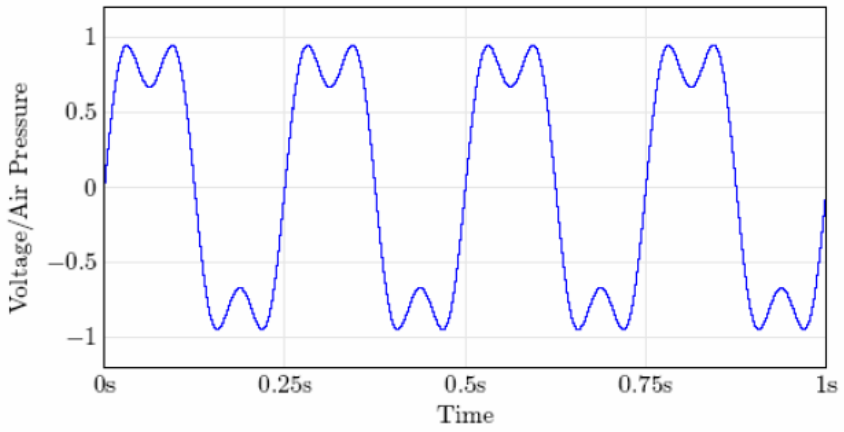


Figure 2.4: 4Hz + 12 Hz sine wave

Notice that the waveform in Figure 2.4 is clearly non-sinusoidal. This example demonstrates the use of a series of sine waves to represent a non-sinusoidal, or complex signal. The next sections are going to develop *Fourier analysis*, which is a mathematical technique that is used to determine the amplitude and frequency of the sinusoidal components contained in any complex signal. Square waves, sawtooth waves, and other periodic signals are analyzed using *Fourier series*. Single pulses and other non-periodic signals are analyzed using *Fourier transforms*.

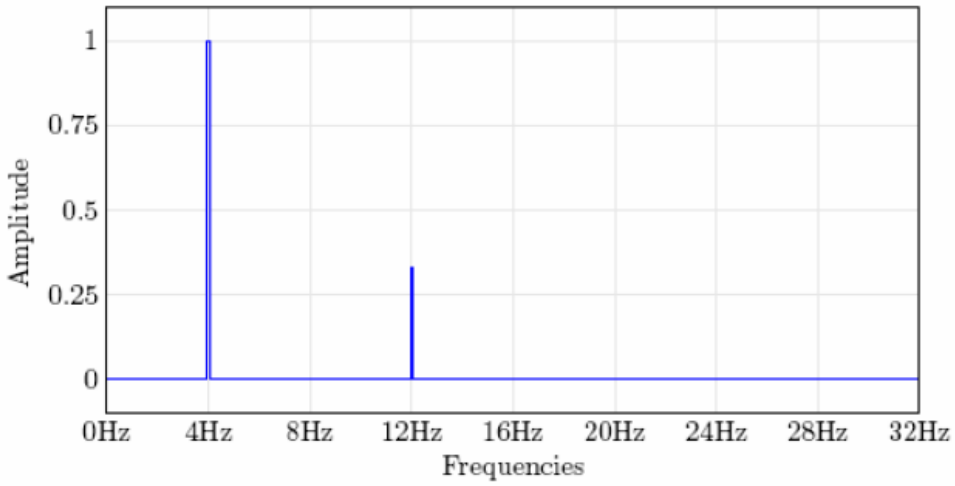


Figure 2.5: Frequency Domain of 4Hz + 12 Hz sine wave

2.3 Fourier Series

Any well-behaved periodic waveform can be represented as a series of sine and/or cosine waves at multiples of its fundamental frequency plus (sometimes) a dc off-set. This is known as a *Fourier series*. Not all signals used in communications are strictly periodic, but they are often close enough for practical purposes.

Fourier's discovery, applied to a time-varying signal $f(t)$, can be expressed mathematically as follows:

$$f(t) = \frac{A_0}{T} + \frac{2}{T}\sum_{n=1}^{\infty}\left(A_n \cos(n\omega t) + B_n \sin(n\omega t)\right) \tag{2.9}$$

where $\omega = 2\pi f_0$, ($f_0 = \frac{1}{T}$ is the *fundamental* frequency), A_0, A_n and B_n are constants calculated from the following relationships:

$$A_0 = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)dt \tag{2.10}$$

$$A_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\cos(n\omega t)dt \tag{2.11}$$

$$B_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\sin(n\omega t)dt \tag{2.12}$$

The A_0 term is called the *DC component* (or dc offset) because it represents the average value of $f(t)$. The A_1 and B_1 terms are *fundamental* components because their period is the same as the period of the complex signal. The remaining terms are *harmonics* because their frequencies are multiples of the fundamental frequency. A_2 and B_2 are the second harmonics, A_3 and B_3 are third harmonics, and so on. The values of the coefficients are calculated by substituting the value of an actual function, $f(t)$, into Equations 2.10 - 2.12 and performing integration.

In some of the waveforms to be encountered, the Fourier coefficients may become zero after integration. Finding zero coefficients is time-consuming and can be avoided. With knowledge of even and odd functions, a zero coefficient may be predicted without performing the integration.

2.3.1 Odd and Even functions

The waveform $f(t)$ is defined as an *even* function if $f(t) = f(-t)$; it has the property of being symmetrical about $t = 0$ axis. If $f(t) = -f(-t)$, the waveform is an *odd* function and has skew symmetry about $t = 0$ axis. A function that has no symmetry about the $t = 0$ axis is neither odd nor even; the sawtooth waveform of Figure 2.6(c) is an example of such a waveform.

Odd and even functions have properties that may be used to simplify Fourier analysis, for example, if

$$f(t) = \frac{A_0}{T} + \frac{2}{T}\sum_{n=1}^{\infty}\left(A_n \cos(n\omega t) \; + B_n \sin(n\omega t)\right) \tag{2.13}$$

then

$$f(-t) = \frac{A_0}{T} + \frac{2}{T}\sum_{n=1}^{\infty}\left(A_n \cos(n\omega t) - B_n \sin(n\omega t)\right) \tag{2.14}$$

If $f(t) = f(-t)$, this can only be true if $B_n = 0$, i.e. the Fourier series of an even function has cosine terms only. Alternatively, if $f(t) = -f(-t)$, then $A_n = 0$, i.e. the Fourier series of an odd function contains only sine terms. If $f(t)$ has no symmetry about $t = 0$, the Fourier series contains both sines and cosines.

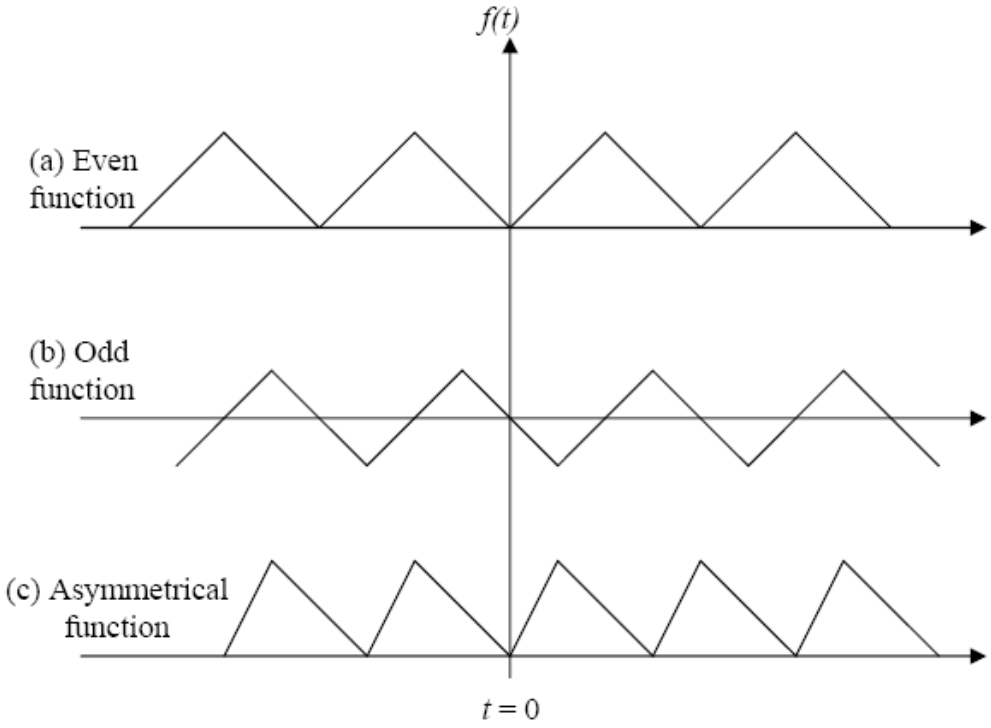


Figure 2.6: Examples of periodic functions

2.3.2 Square wave

A square wave provides an excellent example of the analysis of a periodic wave. A typical time domain display of an ideal square wave is given in Figure 2.7.

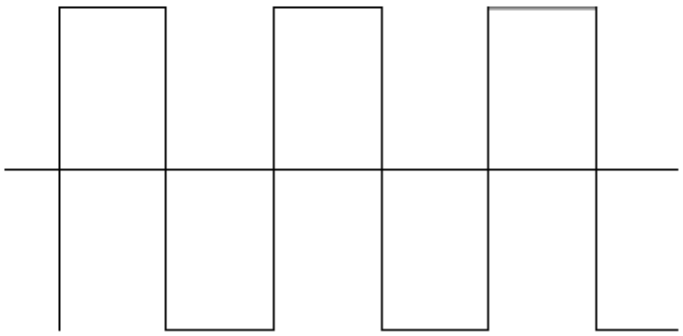


Figure 2.7: Ideal Square wave

Example: Find and sketch the Fourier series corresponding to the odd square wave shown in Figure 2.8.

***Solution*:** This square wave has an amplitude of $V = 1$ V and period of $T = 1$ ms, which implies that its fundamental frequency is $f_0 = 1/T = 1$ kHz. Since this is an odd function, its Fourier series contains only the sine terms given by:

$$B_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\sin(n\omega t)\,dt = -\int_{-\frac{T}{2}}^{0} \sin(n\omega t)\,dt + \int_{0}^{\frac{T}{2}} \sin(n\omega t)\,dt = \frac{1}{n\pi f}\left[1 - \cos(n\pi)\right] \qquad (2.15)$$

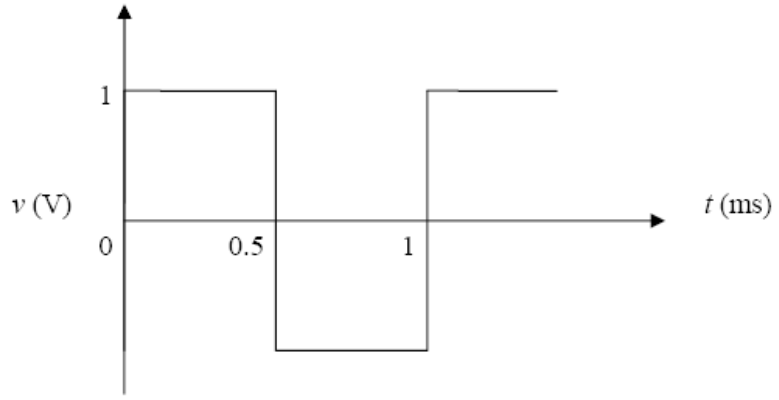


Figure 2.8: Odd Square wave

From the equation above, it can be seen that $B_1 = \frac{2}{\pi f}$, $B_3 = \frac{2}{3\pi f}$, $B_5 = \frac{2}{5\pi f}$, and so on. However, $B_2 = B_4 = B_6 = \ldots = 0$. Hence this odd square wave has components that are all odd multiples of the fundamental frequency of 1 kHz. Substituting the values of B_1, B_3, etc, into equation 2.5, it can be seen that the amplitude decreases with frequency as shown by the following expression:

$$f(-t) = \frac{4}{\pi}\left(\sin \omega t + \frac{1}{3}\sin 3\omega t + \frac{1}{5}\sin 5\omega t + \ldots\right) \tag{2.16}$$

where $\omega = 2\pi \times 10^3$. The 1 kHz component has a peak amplitude of $\frac{4}{\pi} = 1.27$ V, the component at three times the fundamental (3 kHz) has an amplitude one-third that of the fundamental ($\frac{4}{3\pi} = 0.424$ V), that at 5 kHz has an amplitude one-fifth that of the fundamental ($\frac{4}{5\pi} = 0.255$ V), and so on.

Waveform synthesis

This can be regarded as the inverse of Fourier analysis. In effect, the Fourier series indicated that any periodic waveform can be synthesized by adding an infinite number of sinusoidal waves with specific amplitudes and frequencies. In most practical cases, a very good approximation to a given periodic waveform can be obtained by truncating the series to only a few terms.

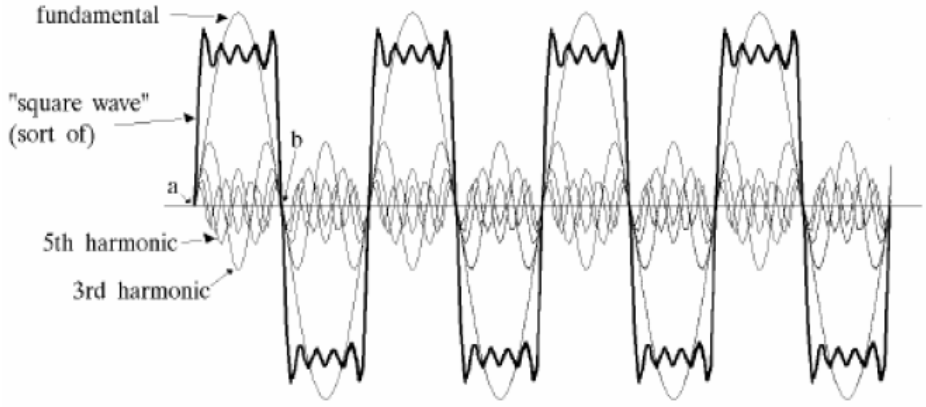


Figure 2.9: Making a square wave out of sine waves

Figure 2.9 shows a fundamental plus four harmonics (the 3rd and 5th are labelled; the 7th and 9th are not), each of which contributes a little to the output. When they are all added up, we get the squarish wave which sort of looks like a square wave, except that its sides do not go straight up and down, and the tops and bottoms are not quite completely flat. This error in waveform synthesis is known as *Gibb's phenomenon*. The reason it only approximates a square wave is that it only has a few odd harmonics, whereas a perfect square wave requires an infinite number of odd harmonics.

For the square wave, the rules are fairly simple:

- The harmonics must be exact odd multiples of the fundamental frequency. For example, the 93rd harmonic of a 1 kHz fundamental would have to be exactly 93 kHz.

- Their amplitude must be just right. For example, the 3rd harmonic must be exactly $\frac{1}{3}$ the size of the fundamental; the fifth harmonic must be exactly $\frac{1}{5}$ the fundamental's size, and so on, all theway up. Thus the 93rd harmonic would have to be $\frac{1}{93}$ the size of the fundamental. This indicates that eventually the harmonics get so small that perhaps they can be omitted without making a noticeable difference in the square wave.

- Their phase has to be just right. To get the steep rise and fall of the square wave, all of the sine waves making it up have to go up together, and down together, as shown at points *a* and *b* in Figure 2.9. If any one of them is out of step, the result will be some other wave, but not a square wave.

2.3.3 Other periodic waves

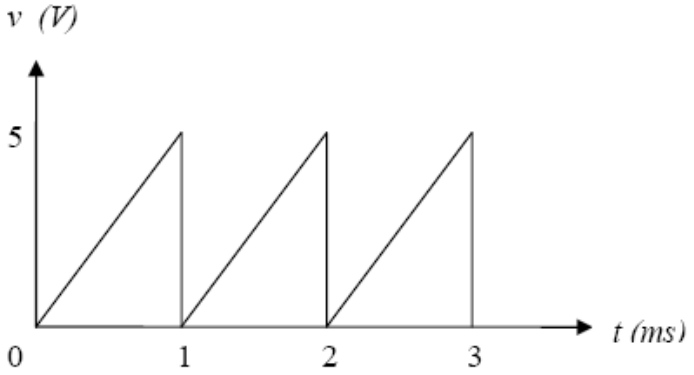


Figure 2.10: Sawtooth in time domain

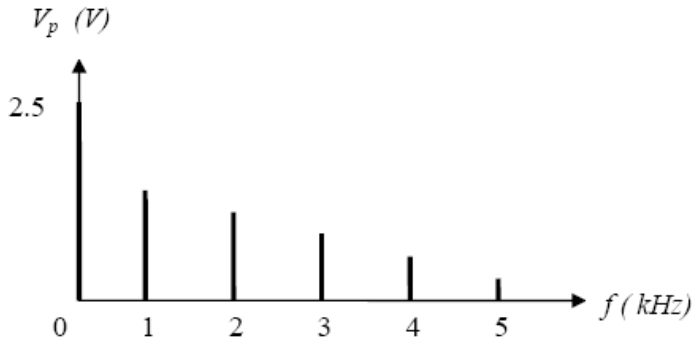


Figure 2.11: Sawtooth in frequency domain

The positive-going sawtooth wave of Figure 2.10 has a Fourier series with a dc term and components at all multiples of the fundamental frequency as shown by the general equation on the table below. The first (dc) term is simply the average value of the signal. For the signal in Figure 2.11, which has a frequency of 1 kHz and a peak amplitude of 5 V, the Fourier series equation becomes:

$$f(t) = 2.5 - 1.59\left\{\sin(2\pi x10^3 t) + 0.5\sin(4\pi x10^3 t) + 0.33\sin(6\pi x10^3 t) + ...\right\} V \qquad (2.17)$$

Periodic Waveform	Symmetry	Fourier Series
Square wave $f(t)=\begin{cases}+V & \lvert t\rvert<\frac{T}{4}\\ -V & \frac{T}{4}\le\lvert t\rvert<\frac{T}{2}\end{cases}$	Even	$f(t)=\frac{4V}{\pi}\left(\cos\omega t-\frac{1}{3}\cos 3\omega t+\frac{1}{5}\cos 5\omega t-\frac{1}{7}\cos 7\omega t+\cdots\right)$
Square wave $f(t)=\begin{cases}-V & -\frac{T}{2}<t<0\\ +V & 0\le t<\frac{T}{2}\end{cases}$	Odd	$f(t)=\frac{4V}{\pi}\left(\sin\omega t+\frac{1}{3}\sin 3\omega t+\frac{1}{5}\sin 5\omega t+\frac{1}{7}\sin 7\omega t+\cdots\right)$
Pulse train $f(t)=\begin{cases}+V & \lvert t\rvert<\frac{\tau}{2}\\ 0 & \frac{\tau}{2}\le\lvert t\rvert<\frac{T}{2}\end{cases}$	Even	$f(t)=\frac{V\tau}{T}+\frac{2V\tau}{T}\left(\frac{\sin\pi\tau/T}{\pi\tau/T}\cos\omega t+\frac{\sin 2\pi\tau/T}{2\pi\tau/T}\cos 2\omega t+\cdots\right)$
Triangle wave $f(t)=V\left(1-4\frac{\lvert t\rvert}{T}\right)\quad \lvert t\rvert<\frac{T}{2}$	Even	$f(t)=\frac{8V}{\pi^2}\left(\cos\omega t+\frac{1}{3^2}\cos 3\omega t+\frac{1}{5^2}\cos 5\omega t+\cdots\right)$
Half-wave Rectified sinusoid $f(t)=\begin{cases}V\sin\omega_o t & 0\le t<\frac{T}{2}\\ 0 & -\frac{T}{2}\le t<0\end{cases}$	None	$f(t)=\frac{V}{\pi}+\frac{V}{2}\sin\omega t-\frac{2V}{\pi}\left(\frac{\cos 2\omega t}{1x3}+\frac{\cos 4\omega t}{3x5}+\cdots\right)$
Full-wave rectified sinusoid (Time zero at voltage zero) $f(t)=\lvert V\sin\omega_o t\rvert$	Even	$f(t)=\frac{2V}{\pi}-\frac{4V}{\pi}\left(\frac{\cos 2\omega t}{1x3}+\frac{\cos 4\omega t}{3x5}+\frac{\cos 6\omega t}{5x7}+\cdots\right)$
Full-wave rectified sinusoid (Time zero at voltage peak) $f(t)=\left\lvert V\sin\omega_o t+\frac{\pi}{2}\right\rvert$	Even	$f(t)=\frac{2V}{\pi}\left(1+\frac{2\cos 2\omega t}{1x3}-\frac{2\cos 4\omega t}{3x5}+\frac{2\cos 6\omega t}{5x7}-\cdots\right)$
Sawtooth (with no dc offset) $f(t)=2V\frac{t}{T}\qquad \lvert t\rvert<\frac{T}{2}$	Odd	$f(t)=\frac{2V}{\pi}\left(\sin\omega t-\frac{1}{2}\sin 2\omega t+\frac{1}{3}\sin 3\omega t-\frac{1}{4}\sin 4\omega t+\cdots\right)$
Sawtooth (positive-going) $f(t)=V\frac{\lvert t\rvert}{T}\qquad \lvert t\rvert<\frac{T}{2}$	None	$f(t)=\frac{V}{2}-\frac{V}{\pi}\left(\sin\omega t+\frac{1}{2}\sin 2\omega t+\frac{1}{3}\sin 3\omega t+\cdots\right)$

Figure 2.12: Fourier-Series for selected waveforms

Example: What bandwidth is required to transmit the first seven components of a triangle wave with a fundamental frequency of 3 kHz? *Solution*: 39 kHz

2.4 Fourier Transform

The Fourier series representation of $f(t)$ is only valid when $f(t)$ is periodic. We have already indicated that information bearing signals change in a random fashion and do not therefore belong to this category. The amplitude spectra of *non-periodic* signals are obtained from the *Fourier transform*. The Fourier transform may be developed from the exponential form of the Fourier series (given in Equation 2.18) by allowing the period T to approach infinity. For instance, a pulse with $T \to \infty$ means that $f(t)$ becomes a pulse of width τ seconds.

$$f(t) = \frac{1}{T}\sum_{-\infty}^{\infty} C_n \exp\left(j2\pi n f_0 t\right) \tag{2.18}$$

where $f_0 = 1/T$ and

$$C_n = \int_{-T/2}^{T/2} f(t)\exp\left(-j2\pi n f_0 t\right)dt \tag{2.19}$$

If we let Δf be the spacing between harmonics in the Fourier series then $\Delta f = (n+1)f_0 - nf_0 = 1/T$. The Fourier series may thus be written as

$$f(t) = \sum_{-\infty}^{\infty} C_n \exp\left(j2\pi n f_0 t\right)\Delta f \tag{2.20}$$

As $T \to \infty$ then $\Delta f \to 0$ and the discrete harmonics in the series merge, and an amplitude spectrum that is a continuous function of frequency results, i.e.

$$\lim_{T\to\infty} C_n = F(f) \tag{2.21}$$

The harmonic number n now has all possible values and the summation of the series can thus be replaced by an integral, i.e. nf_0 is replaced by a continuous function f and

$$f(t) = \int_{-\infty}^{\infty} F(f)\exp\left(j2\pi n f_0 t\right)df \tag{2.22}$$

$$F(f) = \int_{-\infty}^{\infty} f(t)\exp\left(-j2\pi n f_0 t\right)dt \tag{2.23}$$

These two integrals are known as the *Fourier transform pair*. To illustrate the use of the Fourier transform, assume $f(t)$ is a single pulse of amplitude A and duration τ seconds.

$$F(f) = \int_{-\infty}^{\infty} f(t)\exp\left(-j2\pi n f_0 t\right)dt = \int_{-T/2}^{T/2} A\exp\left(-j2\pi f t\right)dt = \frac{A}{\pi f}\sin\left(\pi f\tau\right) \tag{2.24}$$

The *sinc* function , also called the *sampling function*, is a function that arises frequently in signal processing and the theory of Fourier transforms. The full name of the function is *sine cardinal*, but it is commonly referred to by its abbreviation. The sinc function is defined as $sinc(x) = \frac{\sin x}{x}$ and its example plot is shown in Figure 2.13. Observe that the amplitude of the sinc function oscillates, decaying in either direction of x and approaches zero as $|x| \to \infty$. The maximum value of this function occurs as x approaches zero, for $[\sin(x)/x] \to 0$ as $x \to 0$. The latter can be seen by using L'Hopital's rule.

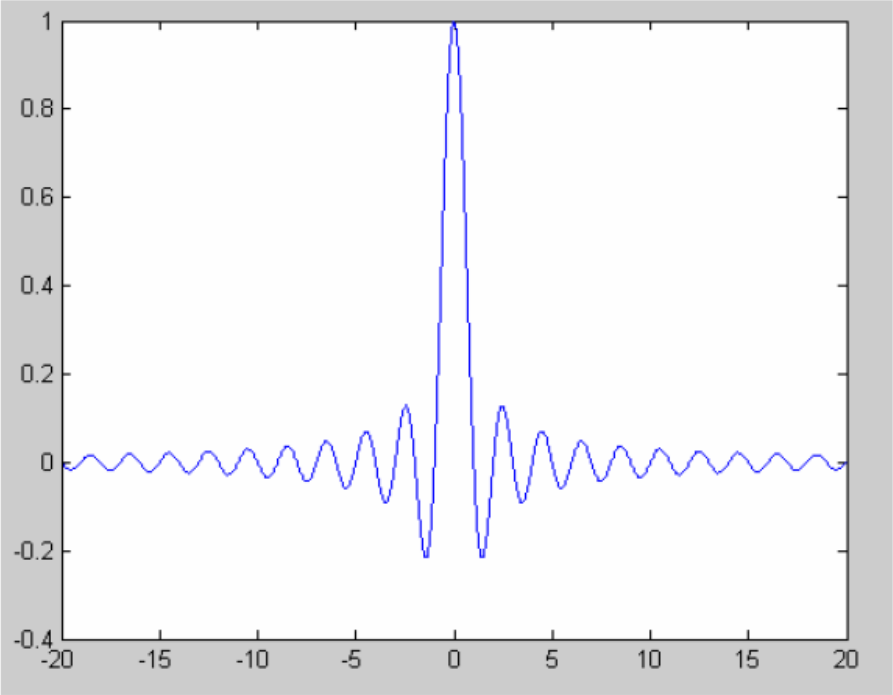


Figure 2.13: An example of a "sinc" function

Thus, the amplitude spectrum from Equation 2.20 is often given as

$$F(f) = A\tau\, sinc(\pi f\tau) \tag{2.25}$$

It should be noted that the envelope of this spectrum is a sinc function that has unity value when $\pi f\tau = 0$ and zero value when $\pi f\tau = m\pi$ i.e. when $f = m/\tau$. A rectangular pulse in the time domain has a Fourier transform that is a sinc function in the frequency domain. The converse is also true, i.e. a sinc pulse in the time domain has a Fourier transform that is a rectangular function in the frequency domain.

Chapter 3

Analog Modulation Schemes

3.1 Introduction

Modulation is the process by which an information carrying signal (often known as baseband or message signal) is shifted to another part of the frequency spectrum. A basic analog communication signal is a carrier signal $v_c(t)$ which is modulated by an information signal $v_m(t)$. A sinusoidal signal is invariably used and such a carrier provides frequency translation of the modulating signal to that part of the radio spectrum where electromagnetic propagation is possible.

We define a sinusoidal carrier signal $v_c(t)$ as:

$$v_c(t) = V_c \cos(\omega_c t + \emptyset_c) \tag{3.1}$$

where V_c = carrier amplitude, ω_c = angular frequency in rad/s and ϕ_c = phase. Note that f_c = carrier frequency (Hz) = $\omega_c/2\pi$ The argument of the cosine function ($\omega_c t + \phi_c$) is also known as the carrier angle (Θ).

When sinusoidal modulation is employed, one of the parameters of $v_c(t)$ is varied in sympathy with the modulating signal $v_m(t)$.

- The amplitude V_c of the carrier may be varied; this is *amplitude modulation* (AM).
- The frequency $\omega_c/2\pi$ may be varied; this is *frequency modulation* (FM).
- The phase ϕ_c can be varied to give *phase modulation* (PM).

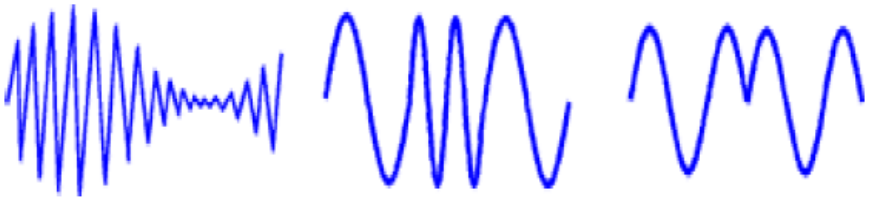


Figure 3.1: Analog Modulation Formats

It will be seen later in the chapter that FM and PM are very similar processes and they are often jointly referred to as *angle modulation*. We shall now consider various modulation methods.

3.2 Amplitude Modulation

For amplitude modulation (also known as *envelope modulation*), the amplitude is made to vary in proportion to the message, $V_c \propto v_m(t)$, and the angle remains constant. In its most general form, amplitude modulated $v_c(t)$ then becomes:

$$V_{am}(t) = [kv_m(t) + C]\cos(\omega_c t + \emptyset_c) \tag{3.2}$$

Amplitude modulation is widely used for radio broadcast communication systems and takes on several forms:-

- Double Sideband with Carrier (DSBC or DSB-WC or DSB-LC)
- Double Sideband with Suppressed Carrier (DSB-SC)
- Single Sideband with Suppressed Carrier (SSB-SC or SSB)
- Vestigial Sideband (VSB)

3.3 DSBC Amplitude Modulation

For DSBC, $k = 1$, $C = V_c$ and $\phi_c = 0$. Then equation 3.2 becomes

$$V_{am}(t) = [\upsilon_m(t) + V_c] \cos \omega_c t \tag{3.3}$$

and for a sinusoidal modulating signal of frequency f_m and amplitude V_m, i.e.

$$\upsilon_m(t) = V_m \cos \omega_m t \tag{3.4}$$

$$V_{am}(t) = [V_m cos\omega_c t + V_c]\, cos\omega_c t = V_c\left[1 + \frac{V_m}{V_c} cos\omega_m t\right] cos\omega_c t = V_c[1 + mcos\omega_m t]cos\omega_c t \tag{3.5}$$

where m is a measure of the degree of modulation, *modulation index* or the depth of modulation, $m = V_m/V_c$. Using the trigonometrical equations given at the end of the chapter,

$$V_{am}(t) = V_c\, cos\omega_c t + mV_m cos\omega_m t.\, cos\omega_c t = V_c cos\omega_c t + \frac{mV_c}{2}\cos(\omega_c + \omega_m)\, t + \frac{mV_c}{2}\cos(\omega_c - \omega_m)t \tag{3.6}$$

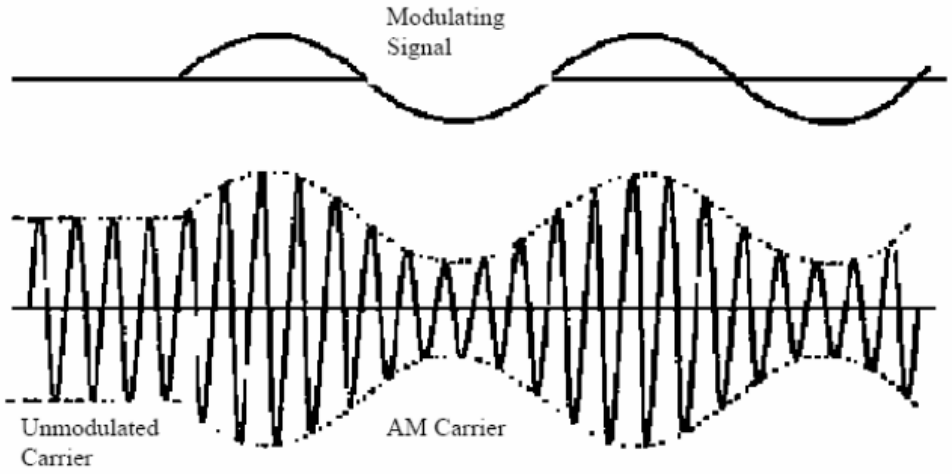


Figure 3.2: A DSBC amplitude modulated signal

This mathematical expression for the AM carrier shows that it is the sum of three sinusoids of different frequencies. One of the sinusoids has the same frequency and amplitude as the unmodulated carrier. The second sinusoid is at a frequency equal to the sum of the carrier frequency and the modulating frequency; this component is the *upper sideband* (USB). The third sinusoid is at a frequency equal to the carrier frequency minus the modulating frequency; this component is the *lower sideband* (LSB). In fact, all the three terms are carriers at different frequencies but only the sideband terms contain the information components V_m and ω_m. It appears that the first term is redundant but, as we shall see later, the retention of this term allows a simple receiver to be adopted. The frequency spectrum for the DSBC can be sketched as shown in Figure 3.3.

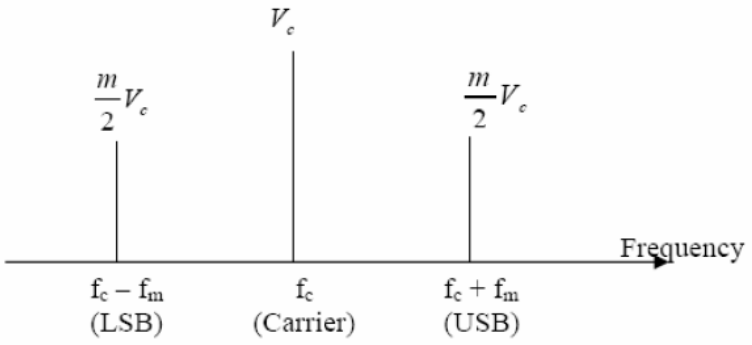


Figure 3.3: DSBC Spectrum

In practice, m will have a value lying between 0 and 1. The envelope modulation produces a waveform which contains higher frequencies than were present in the unmodulated waveform. The overall

bandwidth of the modulated signal is the difference between the highest and the lowest frequencies, namely

$$B = Bandwidth = f_{usb} - f_{lsb} = (f_c + f_m) - (f_c - f_m) = 2f_m \tag{3.7}$$

Modulation index is usually expressed as a percentage called percent modulation. In the time domain, the degree of modulation for a sinusoidal modulation is calculated as (refer to Figure 3.2)

$$m = \frac{V_m}{V_c} = \frac{V_{max} - V_c}{V_c} \tag{3.8}$$

Since the modulation is symmetrical,

$$V_{max} - V_c = V_c - V_{min} \Rightarrow V_c = \frac{V_{max} + V_{min}}{2} \tag{3.9}$$

Therefore, for sinusoidal modulation,

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{3.10}$$

So, it is quite easy to find m by displaying the envelope on an oscilloscope and measuring the maximum and minimum peak-to-peak values for the envelope voltage. All the three components of the modulated signal, when are in phase, add together linearly and form the maximum signal amplitude $V_{max} = V_c + V_{usb} + V_{lsb}$. With the amplitudes of the lower and the upper sidebands being equal, let $V_{SB} = V_{lsb} = V_{usb}$. Therefore,

$$m = \frac{2V_{SB}}{V_c} \tag{3.11}$$

It is worthwhile examining what happens to Equation 3.6 and to the modulated waveform as m varies. To start with, when $m = 0$, $V_m = 0$ and we have the original, unmodulated carrier. As m varies between 0 and 1, the changes due to modulation become more pronounced. Note especially the result for $m = 1$ ($m = 100\%$) under which the peak signal voltage will vary between zero and twice the unmodulated carrier amplitude (or $V_m = V_c$). When the modulation index is greater than 1, *overmodulation* is said to be present and the envelope no longer resembles the modulating signal. Therefore, m must be kept less than or equal to 1.

Example: (a) A 1 MHz carrier with an amplitude of 1 V peak is modulated by a 1 kHz signal with $m = 0.5$. Sketch the amplitude spectrum. (b) An additional 2 kHz signal modulates the carrier with $m = 0.2$. Sketch the spectrum. What is the bandwidth in each case?

Solution: (a) There will be three frequency components at $f_c = 1$ MHz, $f_{usb} = f_c + f_m = 1.001$ MHz and $f_{lsb} = f_c - f_m = 0.999$ MHz. Next we have to determine the amplitudes of these components. The carrier remains is unchanged, so it remains at 1 V peak. The two sidebands will have the same peak voltage $V_{sb} = mV_c/2 = 0.25$ V. $BW = 2f_m = 2$ kHz.

(b) The addition of another modulating signal (f_{m_2}) at a different frequency simply adds another set of side frequencies. The new frequency components will be at 1.002 and 0.998 MHz, and their amplitude will be 0.1 V. $BW = 2f_{mmax} = 4$ kHz.

3.3.1 Power in DSBC AM

A DSBC AM signal can be expressed as shown in Equation 3.6. For this signal, the power delivered to a resistance R is:

$$P_{am} = \frac{V_c^2}{2R} + \frac{m^2 V_c^2}{4R} \tag{3.12}$$

This can also be written as

$$P_{am} = P_c + P_{SB} = P_c\left(1 + \frac{m^2}{2}\right) \quad (3.13)$$

The power in the carrier signal can also be expressed as:

$$P_c = \frac{2}{2+m^2} P_{am} \quad (3.14)$$

while the total power in the sidebands is given by:

$$P_{SB} = \frac{m^2}{2} P_c = \frac{m^2}{2+m^2} P_{am} \quad (3.15)$$

The above equations indicate that for m = 1, the total sideband power in a DSBC signal is 50% of the carrier power, and that as m reduces, the power in the information sidebands reduces rapidly as a square law. In practice, m has to be as large as possible and it is typically 0.6. The transmission efficiency is described as:

$$\eta = \frac{P_{SB}}{P_{am}} = \frac{m^2}{2+m^2} 100\% \quad (3.16)$$

For m = 1, η = 33.3% i.e. only 1/3 of the total power available in the sidebands and the carrier, which does not carry any information, takes up 2/3 of the total transmitted power.

Exercise 1: A DSBC system has a carrier frequency of 470 MHz and a modulating signal frequency of 5 kHz. Calculate the bandwidth of the transmitted signal. If V_c = 24 V, calculate the modulated carrier power for a load of 500 Ω when (a) m = 0, (b) m = 0.5, (c) m = 1.

Solution: ???

3.4 DSB-SC Amplitude Modulation

Although full-carrier AM is simple, it is not a particularly efficient form of modulation in terms of bandwidth. We have seen that the bandwidth is twice the highest modulating frequency because there are two sidebands containing the same information.

Double sideband with suppressed carrier amplitude modulation is achieved when k = 1, C = 0 and ϕ_c = 0 in Equation 3.2. Thus

$$V_{am}(t) = V_m(t)cos\omega_c t = mV_c cos\omega_m t.cos\omega_c t = \frac{mV_c}{2}\cos(\omega_c + \omega_m)\,t + \frac{mV_c}{2}\cos(\omega_c - \omega_m)t \quad (3.17)$$

3.4.1 Power in DSB-SC AM

It has been noticed that two-thirds or more of the transmitted power in DSBC is found in the carrier, which contains no information and merely serve as an aid to demodulation. Removing the carrier from a fully modulated AM signal would change the power available for the sidebands from one-third of a total to all of it.

The power generated in a resistance R by the DSB-SC signal is:

$$P_{am} = \frac{m^2 V_c^2}{4R} = \frac{m^2}{2} P_c = P_{SB} \quad (3.18)$$

Exercise 2: Repeat exercise 1 for a DSB-SC system and compare your results with those for the DSBC case.

Solution: ???

Example: Figure 3.4 shows the effect of removing the carrier from a fully modulated AM signal in the frequency domain. Figure 3.4(a) shows a 1 MHz carrier 100% modulated by a 1 kHz sine wave. The carrier power is 1 W (30 dBm) and each sideband has one-quarter the carrier power (0.25 W), or 24 dBm, giving a total power of 1.5 W. In Figure 3.4(b), we see the result of using the same total signal power (1.5 W) to produce a double-sideband suppressed carrier signal. Since there is no carrier, each sideband has half the total power: 0.75 W (or 28.8 dBm). This shows a power advantage of almost 5 dB (DSB-SC over DSBC).

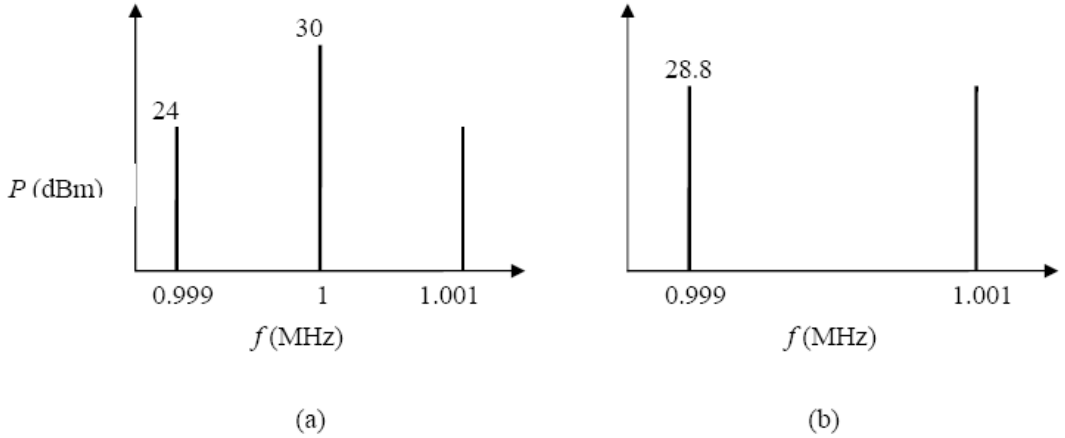


Figure 3.4: Full-carrier AM vs DSB-SC AM

In the time domain, the envelope of the signal will no longer be a faithful representation of the modulating signal. In fact it is merely the sum of the upper and lower sideband signals. DSB-SC AM is not often found on its own as a modulation scheme. It is used as the basis for generating single-sideband suppressed carrier (SSB-SC, or just SSB) signals, which will be discussed in the next section.

3.5 SSB-SC Amplitude Modulation

The upper and lower sidebands of the DSBC and DSB-SC systems contain the same information. Thus transmission of both sidebands is not necessary since either sideband contains the information which will allow us to recover the message at the receiver. By suppressing the carrier and eliminating one sideband, we obtain a single sideband suppressed carrier (SSB-SC) system. Equation 3.17 thus becomes

$$V_{am}(t) = \frac{mV_c}{2}\cos(\omega_c + \omega_m)\,t \ldots or \ldots \frac{mV_c}{2}\cos(\omega_c - \omega_m)t \qquad (3.19)$$

for an SSB-SC system with upper or lower sideband, respectively. Note that the bandwidth requirements are at least half those of DSBC and DSB-SC systems. Since the modulating signal rarely extends right down to dc, the bandwidth improvement will usually be greater than two.

Figure 3.5 illustrates this effect. The baseband shown is a voice signal extending over a frequency range from 300 Hz to 3 kHz. If this signal is transmitted by DSB-SC AM with a carrier frequency of 1 MHz, the bandwidth will be $B = 2f_{m_{max}} = 2 \times 3$ kHz = 6 kHz. With SSB transmission, the bandwidth of one sideband is $B = f_{m_{max}} - f_{m_{min}} = 3kHz - 0.3kHz = 2.7kHz$. Either upper-sideband (USB) [0.997 - 0.9997 MHz] or lower-sideband (LSB) [1.0003 - 1.003 MHz] transmission can be used, and both are found in practice.

This bandwidth reduction has two benefits. The more obvious one is that the signal takes up less spectrum. This allows twice as many signals to be transmitted in a given spectrum allocation. At least as important, however, is the improvement in the signal-to-noise ratio that can be achieved by reducing the bandwidth. If the bandwidth of the transmitted signal is reduced by 50 %, the receiver bandwidth can be reduced by an equivalent amount. Since noise power is proportional to bandwidth, reducing

the receiver bandwidth by one-half eliminates one-half the noise. Assuming the signal power remains constant, this represents a 3 dB improvement in the signal-to-noise ratio.

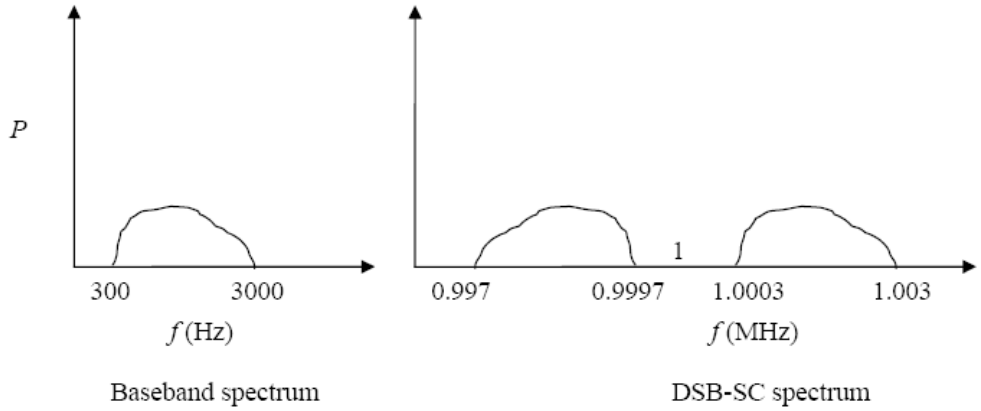


Figure 3.5: SSB transmission of a voice signal

3.5.1 Power in SSB-SC AM

The power generated in R - by the SSB-SC modulated signal is:

$$P_{am} = \frac{m^2 V_c^2}{8R} = \frac{m^2}{4} P_c = \frac{1}{2} P_{SB} \tag{3.20}$$

Exercise 3: Repeat exercise 1 for an SSB-SC system and compare your results with those for the DSBC and DSB-SC cases.

Solution: ???

3.5.2 VSB Amplitude Modulation

This is used for wideband signals, such as television, where the bandwidth of the modulating signal can extend from dc up to 5.5 MHz. The required DSBC AM transmission would therefore be 11 MHz. This is regarded as excessive both from the point of view of transmission bandwidth occupation and of cost. It is generally accepted that the wider the bandwidth of a receiver, the greater the cost.

Since the amplitude spectrum of a video waveform has a dc component, it would be extremely difficult to produce SSB-AM for a television signal. As a compromise, part (i.e. the vestige) of one sideband and the whole of the other is transmitted. A typical example is shown in Figure 3.6.

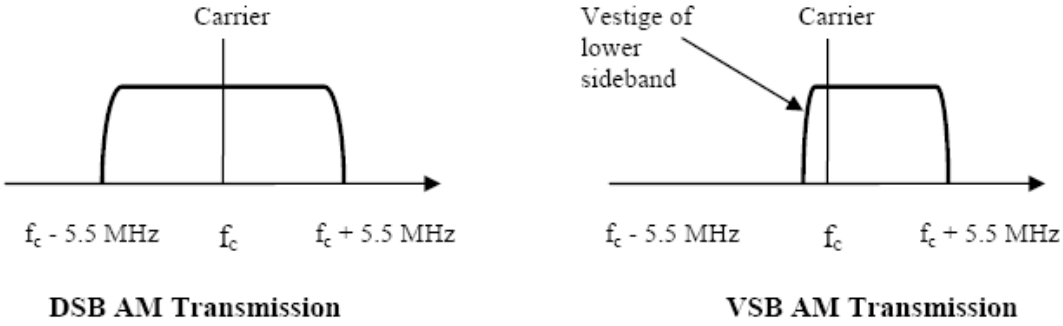


Figure 3.6: Amplitude spectrum of a television signal

The VSB-AM signal thus has both lower power and less bandwidth than DSBC / DSB-SC AM and higher power and greater bandwidth than SSB-AM. The VSB-AM signal, however, does permit a much simpler receiver than a SSB-AM signal.

3.6 Angle Modulation

The carrier $v_c(t) = V_c \cos(\omega_c t + \phi_c)$ has an instantaneous value determined by amplitude V_c and angle $\Theta(t) = \omega_c t + \phi_c$. Angle modulation includes the cases of frequency modulation (FM) and phase modulation. The difference between FM and PM is related to how the instantaneous value of the angle $\Theta(t)$ is varied by the modulating signal $v_m(t)$.

For an unmodulated carrier, the angle is

$$\Theta(t) = \omega_c t + \phi_c \tag{3.21}$$

and if ϕ_c is made instantaneously proportional to $v_m(t)$, then we have *phase modulation* in which

$$\Theta(t) = \omega_c t + K_p v_m(t) \tag{3.22}$$

and K_p is a constant and $K_p v_m(t)$ is the phase deviation.

For frequency modulation, the instantaneous frequency of the modulated carrier has a component proportional to $v_m(t)$, and the instantaneous frequency is

$$\omega(t) = \frac{d\Theta(t)}{dt} = \omega_c + K_f v_m(t) \tag{3.23}$$

where K_f is a constant.

Thus, by integrating equation 3.23, we get

$$\Theta(t) = \omega_c t + K_f \int_0^t v_m(t) \tag{3.24}$$

the constant of integration being taken as zero.

A comparison between equations 3.22 and 3.24 shows that they differ in the form of the second term, and we may write for PM

$$v_c(t) = V_c \cos[\omega_c t + K_p v_m(t)] \tag{3.25}$$

and for FM

$$v_c(t) = V_c \cos\left[\omega_c t + K_f \int_0^t v_m(t)\right] \tag{3.26}$$

We can thus obtain FM from PM by integrating the modulating signal.

3.7 Frequency Modulation

Assume a sinusoidal information (modulating) signal

$$v_m(t) = V_m \cos(\omega_m t) \tag{3.27}$$

then, after integration, equation 3.26 becomes

$$v_c(t) = V_c \cos\left[\omega_c t + \frac{K_f}{\omega_m} V_m \sin(\omega_m t)\right] \tag{3.28}$$

The term

$$\delta = \frac{K_f V_m}{2\pi} \tag{3.29}$$

is defined as the *peak carrier deviation*, and the term

$$m = \frac{\delta}{f_m} \tag{3.30}$$

is defined as the *modulation index*. Equation 3.28 can now be written as

$$v_c(t) = V_c \cos[\omega_c t + m\sin(\omega_m t)] \tag{3.31}$$

which can also be written as

$$\frac{v_c(t)}{V_c} = J_0(m)\cos(\omega_c t) + J_1(m)[\cos(\omega_c+\omega_m)t - \cos(\omega_c-\omega_m)t] + J_2(m)[\cos(\omega_c+2\omega_m)t + \cos(\omega_c-2\omega_m)t] + \ldots \tag{3.32}$$

in which the $J_n(m)$ are the *Bessel functions of order n of the first kind* whose numerical values are given in the table of Figure 3.7.

The Bessel table lists the relative amplitudes of the carrier and each pair of side frequencies as a percentage of the amplitude of the unmodulated carrier for each modulation index. The table shows that the number of side frequency pairs, and therefore the bandwidth, increases as the modulation index increases. The relative amplitude of the unmodulated carrier is listed with a value of 1.00. The symbol J_0 represents the carrier, J_1 represents the first pair of side frequencies, J_2 the second pair, etc. Any side frequency amplitudes that are less than one percent of the unmodulated carrier amplitude are not listed in the Bessel table because it is common practice to ignore them. Some of the carrier and side frequency amplitudes have negative values. This means that the frequency represented by that value is shifted by 180 degrees, indicating phase inversion.

Bessel Functions of the First Kind, $J_n(x)$

x	J_0	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8	J_9	J_{10}
0.0	1.00										
0.2	0.99	0.10									
0.4	0.96	0.20	0.02								
0.6	0.91	0.29	0.04								
0.8	0.85	0.37	0.08	0.01							
1.0	0.77	0.44	0.11	0.02							
1.2	0.67	0.50	0.16	0.03	0.01						
1.4	0.57	0.54	0.21	0.05	0.01						
1.6	0.46	0.57	0.26	0.07	0.01						
1.8	0.34	0.58	0.31	0.10	0.02						
2.0	0.22	0.58	0.35	0.13	0.03	0.01					
2.2	0.11	0.56	0.40	0.16	0.05	0.01					
2.4	0.00	0.52	0.43	0.20	0.06	0.02					
2.6	(0.10)	0.47	0.46	0.24	0.08	0.02	0.01				
2.8	(0.19)	0.41	0.48	0.27	0.11	0.03	0.01				
3.0	(0.26)	0.34	0.49	0.31	0.13	0.04	0.01				
3.2	(0.32)	0.26	0.48	0.34	0.16	0.06	0.02				
3.4	(0.36)	0.18	0.47	0.37	0.19	0.07	0.02	0.01			
3.6	(0.39)	0.10	0.44	0.40	0.22	0.09	0.03	0.01			
3.8	(0.40)	0.01	0.41	0.42	0.25	0.11	0.04	0.01			
4.0	(0.40)	(0.07)	0.36	0.43	0.28	0.13	0.05	0.02			
4.2	(0.38)	(0.14)	0.31	0.43	0.31	0.16	0.06	0.02	0.01		
4.4	(0.34)	(0.20)	0.25	0.43	0.34	0.18	0.08	0.03	0.01		
4.6	(0.30)	(0.26)	0.18	0.42	0.36	0.21	0.09	0.03	0.01		
4.8	(0.24)	(0.30)	0.12	0.40	0.38	0.23	0.11	0.04	0.01		
5.0	(0.18)	(0.33)	0.05	0.36	0.39	0.26	0.13	0.05	0.02	0.01	
5.2	(0.11)	(0.34)	(0.02)	0.33	0.40	0.29	0.15	0.07	0.02	0.01	
5.4	(0.04)	(0.35)	(0.09)	0.28	0.40	0.31	0.18	0.08	0.03	0.01	
5.6	0.03	(0.33)	(0.15)	0.23	0.39	0.33	0.20	0.09	0.04	0.01	
5.8	0.09	(0.31)	(0.20)	0.17	0.38	0.35	0.22	0.11	0.50	0.02	0.01
6.0	0.15	(0.28)	(0.24)	0.11	0.36	0.36	0.25	0.13	0.06	0.02	0.01
6.2	0.20	(0.23)	(0.28)	0.05	0.33	0.37	0.27	0.15	0.07	0.03	0.01
6.4	0.24	(0.18)	(0.30)	(0.01)	0.29	0.37	0.29	0.17	0.08	0.03	0.01
6.6	0.27	(0.12)	(0.31)	(0.06)	0.25	0.37	0.31	0.19	0.10	0.04	0.01
6.8	0.29	(0.07)	(0.31)	(0.12)	0.21	0.36	0.33	0.21	0.11	0.05	0.02
7.0	0.30	0.00	(0.30)	(0.17)	0.12	0.35	0.34	0.23	0.13	0.06	0.02
7.2	0.30	0.05	(0.28)	(0.21)	0.11	0.33	0.35	0.25	0.15	0.07	0.03
7.4	0.28	0.11	(0.25)	(0.24)	0.50	0.30	0.35	0.27	0.16	0.08	0.04
7.6	0.25	0.16	(0.21)	(0.27)	0.00	0.27	0.35	0.29	0.18	0.10	0.04
7.8	0.22	0.20	(0.16)	(0.29)	(0.06)	0.23	0.35	0.31	0.20	0.11	0.05
8.0	0.17	0.23	(0.11)	(0.29)	(0.11)	0.19	0.34	0.32	0.22	0.13	0.06
8.2	0.12	0.26	(0.06)	(0.29)	(0.15)	0.14	0.32	0.33	0.24	0.14	0.07
8.4	0.07	0.27	0.00	(0.27)	(0.19)	0.09	0.30	0.34	0.26	0.16	0.08
8.6	0.01	0.27	0.05	(0.25)	(0.22)	0.04	0.27	0.34	0.28	0.18	0.10
8.8	(0.04)	0.26	0.10	(0.22)	(0.25)	(0.01)	0.24	0.34	0.29	0.20	0.11
9.0	(0.09)	0.25	0.14	(0.18)	(0.27)	(0.06)	0.20	0.33	0.31	0.21	0.12
9.2	(0.14)	0.22	0.18	(0.14)	(0.27)	(0.10)	0.16	0.31	0.31	0.23	0.14
9.4	(0.18)	0.18	0.22	(0.09)	(0.27)	(0.14)	0.12	0.30	0.32	0.25	0.16
9.6	(0.21)	0.14	0.24	(0.04)	(0.26)	(0.18)	0.08	0.27	0.32	0.27	0.17
9.8	(0.23)	0.09	0.25	(0.01)	(0.25)	(0.21)	0.03	0.25	0.32	0.28	0.19
10.0	(0.25)	0.04	0.25	0.06	(0.22)	(0.23)	(0.01)	0.22	0.32	0.29	0.21

NB: Numbers in bracket indicate negatives (e.g -0.27 = (0.27))

Figure 3.7: Bessel Table

3.7.1 FM Bandwidth

Theoretically, FM produces an infinite number of side frequencies, causing an infinitely large bandwidth. In practice, side frequencies with amplitudes less than *one percent* (0.01) of the unmodulated carrier amplitude are considered insignificant.

Carson's rule provides a quick way to approximate the bandwidth of a frequency-modulated carrier. It states that the bandwidth (B) is proportional to the maximum modulating frequency (f_m) and the frequency deviation (δ). Therefore,

$$B = 2(f_m + \delta) = 2f_m(1 + m) \tag{3.33}$$

In order to more accurately determine the bandwidth of an FM carrier, the *Bessel table* must be used. The FM modulation index must be first determined in order to use the Bessel table to determine the expected bandwidth. The numbers on the same line as this modulation index represent the normalized values of the carrier amplitude and the side frequency amplitude pairs that are more than one percent of the unmodulated carrier amplitude. The number of side frequency pairs for this modulation index must be counted and recorded as *N*. Using this value of *N* and the modulating signal frequency (f_m), the bandwidth is calculated using the following equation:

$$B = 2f_mN \tag{3.34}$$

This equation is based on the fact that each side frequency is separated by a frequency equal to the modulating signal frequency (f_m) and there are *N* side frequencies on each side of the carrier.

Example: VHF broadcasting uses FM with a maximum frequency deviation of 75 kHz and highest modulation frequency of 15 kHz. Plot the sideband spectrum and calculate the bandwidth required for transmission.

Solution: The modulation index $m = \frac{\delta}{f_m} = \frac{75}{15} = 5$. Using the Bessel table, the values of the Bessel function for $m = 5.0$ are: $J_0 = -0.18$, $J_1 = -0.33$, $J_2 = 0.05$, $J_3 = 0.36$, $J_4 = 0.39$, $J_5 = 0.26$, $J_6 = 0.13$, $J_7 = 0.05$, $J_8 = 0.02$, $J_9 = 0.01$

Substituting the numerical values of the Bessel coefficients in Equation 3.32, the sideband spectrum can be plotted and is shown in Figure 3.8. The nine significant sidebands calculated above, separated by the modulation frequency, give a bandwidth of $B = 2 \times 15 \times 9 = 270$ kHz. Using the approximate Carson's rule, the bandwidth is $B = 2(75 + 15) = 180$ kHz, which underestimates bandwidth slightly.

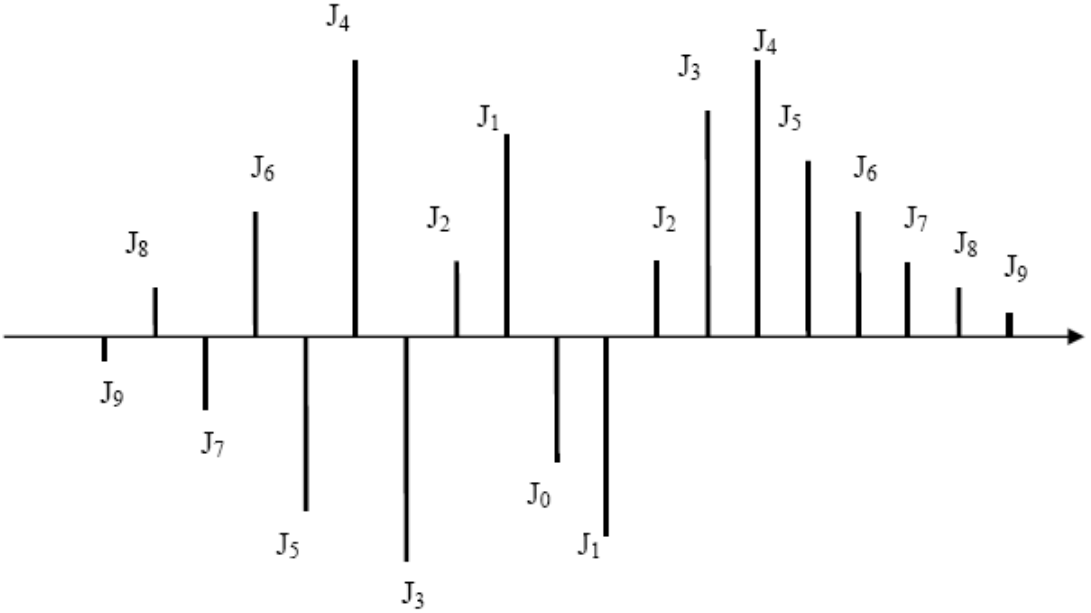


Figure 3.8: Bandwidth Spectrum for m = 5.0

The Bessel table can also be used to determine the modulation index from the frequency spectral plot of the FM carrier. The number of side frequency pairs (N) and the carrier amplitude should be determined from the spectral plot. The modulation index can be determined from these values using the Bessel table. All side frequency pairs on the spectral plot with amplitudes less than one percent of the modulated carrier amplitude should be ignored.

Any FM signal that has only one pair of significant side frequencies is called *narrowband FM* (NBFM). Because NBFM has only two side frequencies, it occupies no more spectral space than AM. The Bessel table verifies that the largest modulation index that generates only one pair of side frequencies is 0.25. Therefore, any frequency-modulated carrier that has a modulation index of 0.25 or less can be considered narrowband FM. Because there are only two side frequencies (i.e. $N = 1$), the bandwidth is determined by multiplying the modulating frequency by two. Therefore,

$$B = 2f_m \tag{3.35}$$

The primary purpose of NBFM is to conserve frequency spectrum, but it does so at the expense of reduced signal-to-noise ratio. Any FM signal with a modulation index greater than 0.25 is considered *wideband FM*.

So far, only single-frequency sine wave modulating signal have been assumed. A complex modulating signal, such as a square wave, consists of a fundamental sine wave frequency and many more harmonics, causing many more modulating frequencies to be generated. In this case, the highest modulating frequency should be used to determine the modulation index and the bandwidth. The higher the modulating signal frequencies (meaning more information is transmitted), the wider the modulated carrier bandwidth. This is the reason a video signal occupies more bandwidth than an audio signal. Because signals transmitted on the same frequency interfere with one another, this increased bandwidth places a limit on the number of modulated carriers that can occupy a given communications channel. Also, when a frequency-modulated carrier is over-modulated, it will have a higher frequency deviation and a higher modulation index, which will cause additional side frequencies to be generated. This causes the transmitted bandwidth to increase and interfere with other signals.

3.7.2 Power in FM

One of the most important differences between FM and AM is the distribution of power in the modulated wave shape. A modulated wave shape is a composite of the carrier and side frequencies. Therefore, the total power transmitted is the sum of the carrier power and the power in the side frequencies. *Unlike AM, the total power in a frequency-modulated carrier is equal to the power of the unmodulated carrier.* Therefore, the power that was originally in the unmodulated carrier is redistributed between the carrier and the side frequencies during modulation. For this reason, the carrier amplitude decreases as the number of side frequencies goes up, as can be seen from the Bessel table. Therefore, the sum of the side frequency amplitudes and the carrier amplitude remains constant for all modulation indexes.

Some of the advantages of FM over AM are:

- *Increased noise immunity* - FM does not change the amplitude of the carrier and this makes FM less sensitive to noise because noise is an amplitude variation.
- *Improved fidelity* - FM has improved fidelity because it introduces less distortion than AM.
- *More efficiency* - FM is more efficient than AM because linear amplifiers are not necessary due to the constant amplitude of FM signals.

Some disadvantages of FM when compared to AM include *requiring more bandwidth* and *requiring more complex circuits in the transmitter and receiver*. Because FM occupies so much bandwidth, it is used only at the higher frequencies of the frequency spectrum, where there is more room.

FM is more familiar in daily life since it is used extensively for radio broadcasting. FM is also used for the sound signal in television, for two-way fixed and mobile radio systems, to name only a few of its more common applications.

3.8 Phase Modulation

Phase modulation (PM) results whenever a baseband information signal varies the phase of a high frequency carrier, with the carrier amplitude remaining constant. However, when the frequency of a carrier is varied, the phase also varies, and when the phase of a carrier is varied, the frequency also varies. Therefore PM is just a different form of FM and the carrier is similar in appearance to an FM.

The role of the modulation index m remains the same as in FM. Thus we can compute the deviation $\delta = mf_m = \Delta\Theta f_m$ where $\Delta\Theta$ is the peak phase deviation equal to the numerical value of m. We can then proceed as if the modulation were FM so far as bandwidth, sidebands, etc, are concerned.

3.9 Trigonometric Identities

$$\sin(A + B) = \sin A\cos B + \cos A\sin B \tag{3.36}$$

$$\sin(A - B) = \sin A\cos B - \cos A\sin B \tag{3.37}$$

$$\cos(A + B) = \cos A\cos B - \sin A\sin B \tag{3.38}$$

$$\cos(A - B) = \cos A\cos B + \sin A\sin B \tag{3.39}$$

$$\sin A\sin B = \frac{1}{2}[cos(A - B) - cos(A + B)] \tag{3.40}$$

$$\cos A\cos B = \frac{1}{2}[cos(A - B) + cos(A + B)] \tag{3.41}$$

$$\sin A\cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)] \tag{3.42}$$

$$\cos^2 A = \frac{1}{2}[1 + \cos 2A] \tag{3.43}$$

$$\sin^2 A = \frac{1}{2}[1 - \cos 2A] \tag{3.44}$$

$$\sin 2A = 2\sin A\cos A \tag{3.45}$$

$$\cos 2A = \cos^2 A - sin^2 A \tag{3.46}$$

$$sin^2 A + cos^2 A = 1 \tag{3.47}$$

$$\cos(m\sin A) = J_0(m) + 2[J_2(m)\cos 2A + J_4(m)\cos 4A + \ldots] \tag{3.48}$$

$$\sin(m\sin A) = 2[J_1(m)\sin A + J_3(m)\sin 3A + \ldots] \tag{3.49}$$

$$\sin(m\cos A) = 2[J_1(m)\cos A - 2J_3(m)\cos 3A + 2J_5(m)\cos 5A + \ldots] \tag{3.50}$$

$$\sin(m\sin A) = 2[J_1(m)\sin A + J_3(m)\sin 3A + \ldots] \tag{3.51}$$

$$\cos(m\cos A) = J_0(m) - 2[J_2(m)\cos 2A + J_4(m)\cos 4A + \ldots] \tag{3.52}$$

Chapter 4

Modulators and Demodulators

4.1 Introduction

In communications, a *mixer* is a nonlinear circuit that combines two signals in such a way as to produce the sum and difference of the two input frequencies at the output. Sometimes the original input frequencies are also present. That is, mixers provide a way of moving a signal from one frequency to another. A*balanced mixer* is one in which the input frequencies do not appear at the output, the only frequencies that are produced are the sum and difference of input frequencies. This is equivalent to the generation of a double-sideband suppressed carrier (DSB-SC) signal.

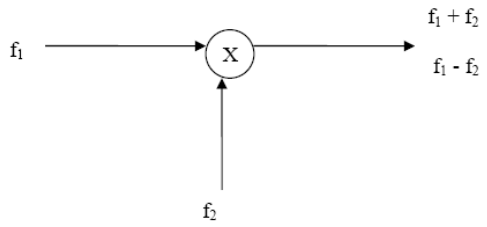


Figure 4.1: A Mixer Symbol

4.2 AM Modulation and Demodulation

Mixing and amplitude modulation are essentially the same process. Both amplitude modulators and ordinary mixers are provided with two input frequencies. They produce the sum and difference signals at their output, as well as the two original signals. In an AM modulator, the two inputs are the baseband and the carrier signals, and the sum and difference signals are, of course, the USB and LSB.

The *demodulator*, also called the *detector*, is the part of the receiver that recovers the baseband signal. It performs the inverse operation to the transmitter modulator. The circuitry used for this varies, of course, with the modulation scheme.

4.2.1 DSBC AM Generation and Detection

DSBC AM is produced by mixing (multiplying) together the carrier and the modulating signal. The mixing is achieved by using a network with a non-linear characteristic. The diode modulator, shown in Figure 4.2, is a good example.

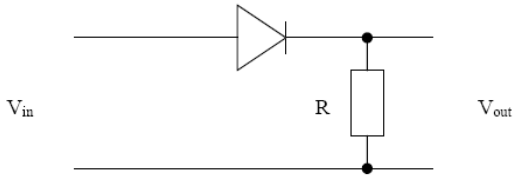


Figure 4.2: DSBC Diode Modulator

The input/output characteristic of the circuit of Figure 4.2 can be written in terms of a power series, i.e.

$$V_{out} = aV_{in} + bV_{in}^2 + cV_{in}^3 + \ldots \quad (4.1)$$

If

$$V_{in} = V_c \cos \omega_c t + V_m \cos \omega_m t \quad (4.2)$$

the output is given by

$$V_{out} = a\left[V_c \cos \omega_c t + V_m \cos \omega_m t\right] + b\left[V_c^2 \cos^2 \omega_c t + V_m^2 \cos^2 \omega_m t + 2V_c V_m \cos \omega_c t \cos \omega_m t\right] + \ldots \quad (4.3)$$

i.e.

$$V_{out} = aV_c \cos \omega_c t[1 + K_1 \cos \omega_m t] + otherterms \quad (4.4)$$

where $K_1 = 2V_m/a$. If the 'other terms' are filtered, Equation 4.4 has the same form as Equation 3.5. Hence the diode produces the required modulation.

The detection (i.e. demodulation) of DSBC AM can be considered broadly under two headings; these are *non-coherent detection* and *coherent (synchronous) detection*. The simplest, and most popular, non-coherent demodulator for full-carrier AM is the envelope detector. It recovers the baseband signal by rectifying the incoming signal to remove half the envelope, and then use a lowpass filter to remove the high-frequency components of the signal. Figure 4.3 shows the operation of an envelope detector. The output of the lowpass filter will be a baseband signal with a dc offset, which can be easily removed with a series capacitor.

Coherent detection operates on the principle that is the inverse of modulation, i.e. the frequency of the sidebands is translated back to baseband by multiplying the DSBC AM signal by a sinusoid of the same frequency as the carrier. In the case of DSBC AM the carrier is actually transmitted, so the problem is one of extracting this component from the DSBC AM signal and then applying the extracted component to a mixer along with the original modulated signal. A block diagram of the coherent detector, using **Phase Locked Loop (PLL)**, is shown in Figure 4.4.

The output of the coherent detector may be written as

$$V_{out} = K[1 + m\cos \omega_m t] \cos \omega_c t \cos \omega_c t \quad (4.5)$$

i.e.

$$V_{out} = K[1 + m\cos \omega_m t] \cos^2 \omega_c t \quad (4.6)$$

but

$$\cos^2 \omega_c t = \frac{1}{2}[1 + \cos 2\omega_c t] \quad (4.7)$$

Therefore

$$V_{out} = \frac{1}{2}Km \cos \omega_m t + unwantedterms \quad (4.8)$$

The unwanted terms will be sidebands of $2\omega_c$ and are easily filtered off. A convenient method of extracting the carrier is to use a phase locked loop as shown in Figure 4.4.

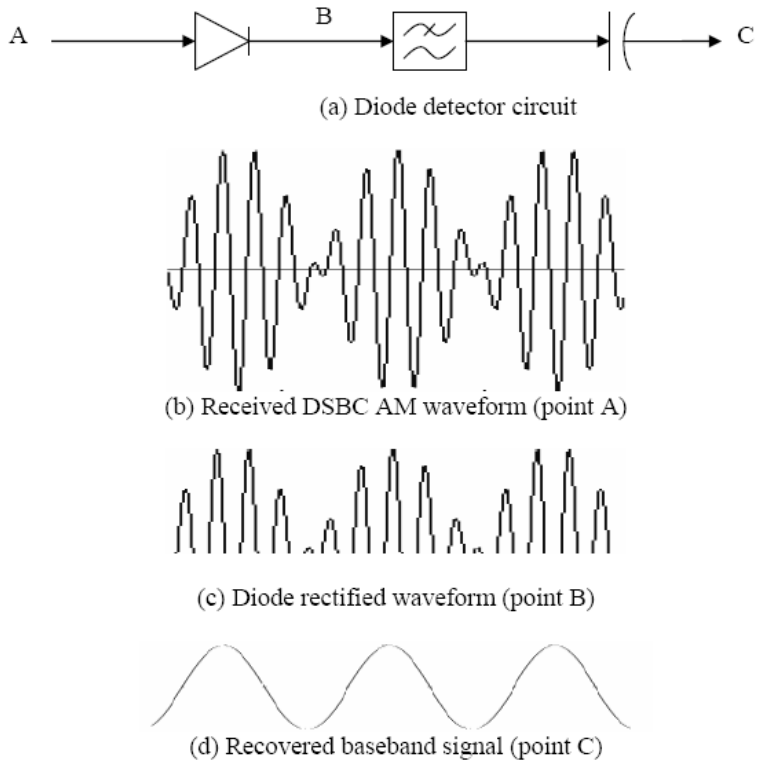


Figure 4.3: Envelope Detection

The basic operation of the loop may be described as follows. The input to the phase detector is the DSBC AM signal and a sinusoid from a voltage controlled oscillator (VCO). The output of the mixer is a signal proportional to the phase difference between the carrier of the DSBC AM signal and the VCO output. This error signal is fed via a lowpass filter to the VCO input. Its effect is to cause the output frequency of the VCO to change in order to minimize the phase error. When the loop is locked, the frequency of the VCO output is equal in frequency to the incoming carrier.

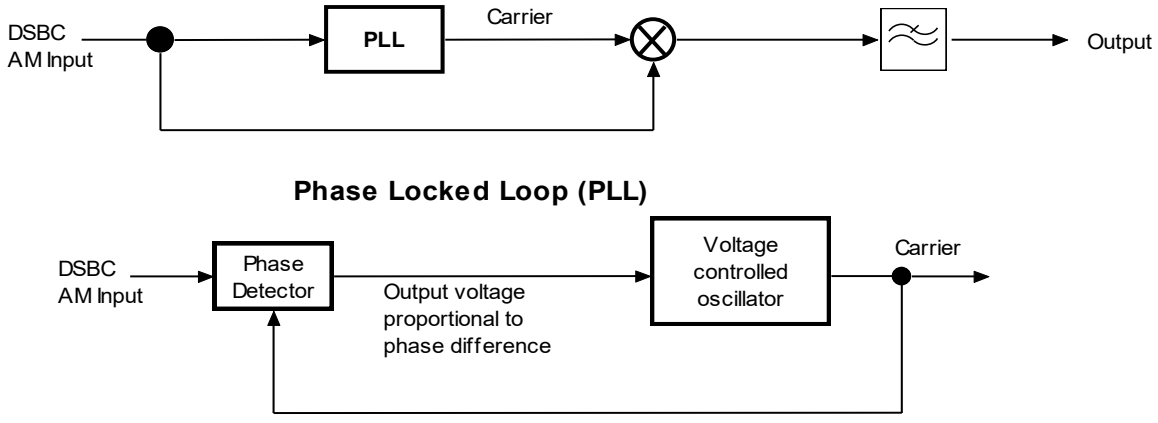


Figure 4.4: Coherent detection of DSBC AM

4.2.2 DSB-SC AM Generation and Detection

Applying the idea of balanced mixers, a balanced modulator would end up with a signal that has upper and lower sidebands, but no carrier; in other words, a DSB-SC signal. Figure 4.5 shows such a modulator fed with sinusoidal carrier and baseband signals (ignoring amplitude factors).

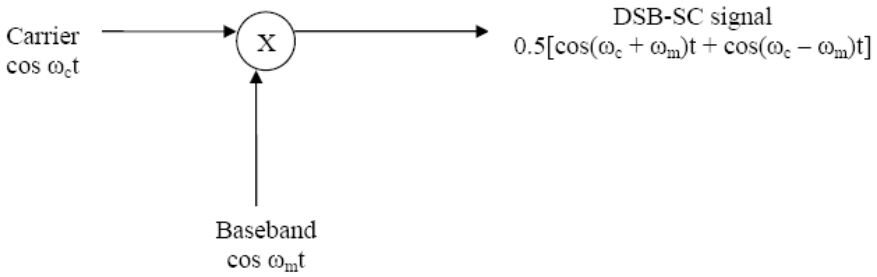


Figure 4.5: DSB-SC Generation

A diode detector used alone will not work for DSB-SC or SSB because the envelope is different from that of DSBC (it does not contain the carrier). This type of modulation requires coherent detection, but since there is no carrier transmitted, the required component must be generated by a local oscillator. Special circuits are required to ensure that the local oscillator is locked in phase to the incoming signal.

A locally generated carrier can be injected along with the DSB-SC signal into a diode detector, but it is better to use an arrangement similar to the one used to generate suppressed-carrier signals (that is, a balanced modulator). When used in receivers, such a device is called a product detector. Consider the detector in Figure 4.6. The two inputs to the detector consist of the signal to be demodulated and a locally generated carrier, produced by a local oscillator.

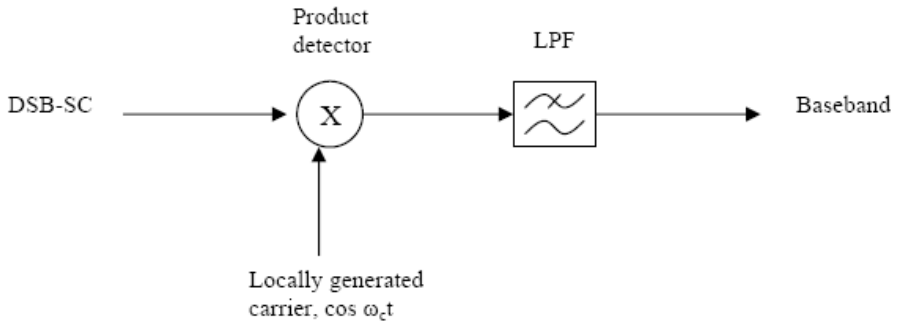


Figure 4.6: DSB-SC Product Detector

Both the correct phase and frequency must be known to correctly demodulate DSB-SC waveforms. Special circuits are required to ensure that the local oscillator is locked in phase to the incoming signal. If the incoming signal is represented by

$$V_{in} = K \cos \omega_c t \cos \omega_m t \tag{4.9}$$

and the output of the local oscillator is $\cos(\omega_c t + \phi)$, the output of the detector will be

$$V_{out} = \frac{1}{2} K \cos \omega_m t [\cos 2\omega_c t + \cos \phi] \tag{4.10}$$

After filtering off the components centred at $2\omega_c$, this gives a term $0.5K \cos \omega_m t \cos \phi$. Thus, if there is a phase error between the original carrier and the local oscillator output, the output varies according to $\cos \phi$. When $\phi = 0$ (i.e. there is no phase error), the output is a maximum and is proportional to the original modulating signal; when $\acute{A} = 90_o$ the output is zero (i.e. the whole signal is wiped out).

4.2.3 SSB-SC AM Generation and Detection

The obvious way to eliminate the unwanted sideband, once the carrier has been suppressed, is to use a bandpass filter. Ideally, simply filtering the carrier would also work. However, in an AM signal, the carrier is both very large in amplitude and very near the wanted sideband. There would be practical difficulties in removing the carrier while leaving the sideband untouched. It is therefore more convenient to null the carrier with a balanced modulator and then remove one of the sidebands by filtering, as shown by the block diagram of Figure 4.7.

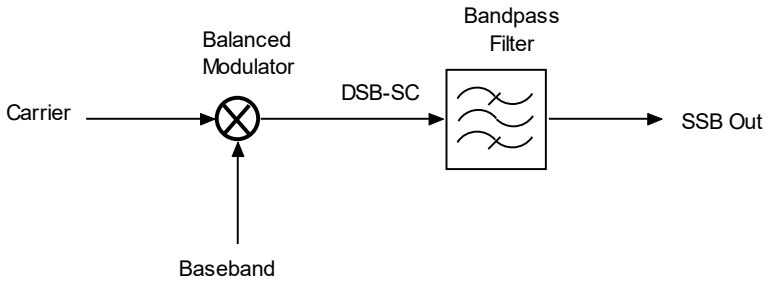
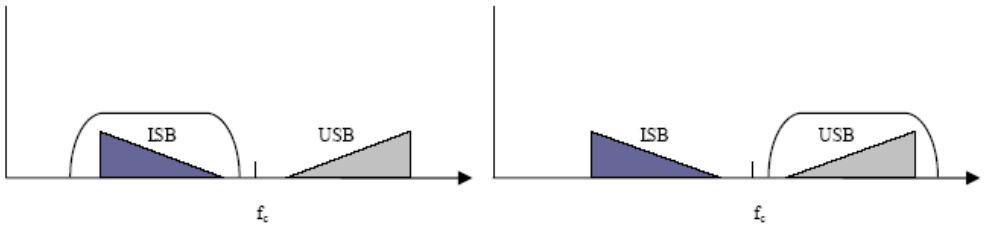


Figure 4.7: SSB-SC Generation

The sidebands are usually shown as triangles, not because that is the shape of the spectrum but in order to emphasize that they are mirror images of each other and to make clear which sideband is present when a single-sideband signal is shown.



(a) LSB Generation (b) USB Generation

Figure 4.8: LSB or USB by filter method

Even with the carrier suppressed, the cut-off of the filter must be quite sharp. The lowest baseband frequency transmitted is likely to be about 300 Hz in a communications-quality voice signal. Thus, there is a gap of 600 Hz between the two sidebands. The attenuation of the filter must change from 0 dB to at least 60 dB within this range.

An interesting alternative method is based on the *Hilbert transformation*. The Hilbert transform of a signal $v(t)$ is the sum of the individual frequency components that have all been shifted by 90^0. The USB SSB AM signal for a single modulating tone is

$$v_{usb}(t) = A\cos[(\omega_c + \omega_m)t] \tag{4.11}$$

i.e.

$$v_{usb}(t) = A[\cos(\omega_c t)\cos(\omega_m t) - \sin(\omega_c t)\sin(\omega_m t)] \tag{4.12}$$

Equation 4.12 reveals that $v_{usb}(t)$ is obtained by subtracting the outputs of two DSB-SC AM modulators, often called balanced modulators. There is a quadrature phase relationship between the inputs of the first modulator relative the inputs of the second modulator. A block diagram of this type of SSB AM modulator is given in Figure 4.9 and it should be noted that sideband filters are not required. The modulator does, however, require the provision for wideband phase-shifting networks.

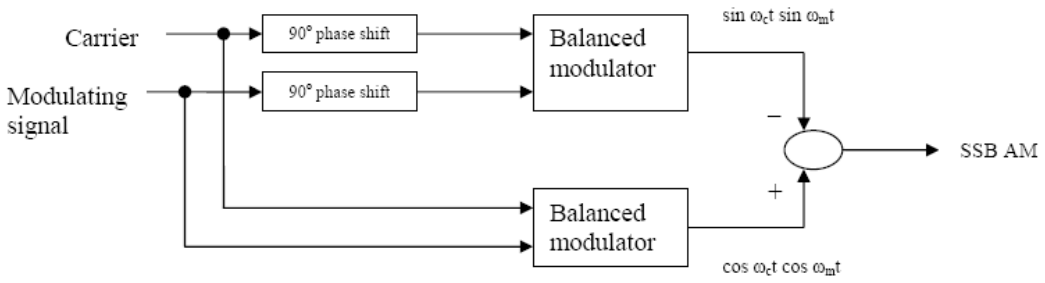


Figure 4.9: Hilbert transform production of SSB AM

SSB requires the use of a synchronous detector for detection similar to DSB-SC, the only difference is that the signal to be demodulated contains only one sideband. Consider an SSB signal with single-tone modulation using the USB. Assuming that this signal was originally produced by modulating a cosine wave carrier at ω_c with another cosine wave at ω_m, it will have the following equation (neglecting amplitude factors):

$$v_{usb} = \cos(\omega_c + \omega_m)t \tag{4.13}$$

Let the locally generated carrier be

$$v_c = \cos\ \omega_c t \tag{4.14}$$

The product detector will multiply these two signals to produce the output

$$v_o = \cos(\omega_c + \omega_m)t \cos\ \omega_c t = 0.5[\cos\ \omega_m t + \cos(2\omega_c + \omega_m)t \tag{4.15}$$

This signal has two components. The first is simply the demodulated baseband signal. The second component can be removed by passing the output signal through a lowpass filter. This is quite easy because the unwanted component has a frequency more than twice the carrier frequency ($2\omega_c + \omega_m$) and therefore is several orders of magnitude higher than the baseband frequency.

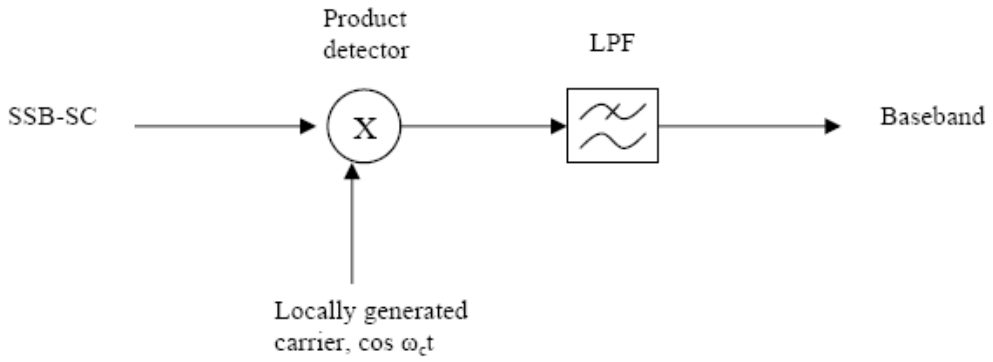


Figure 4.10: SSB Product Detector

Assuming the local oscillator signal is $\cos(\omega_c t + \phi)$, i.e. there is a phase error ϕ, the output of the detector will be

$$V_{out} = \frac{1}{2}K\cos[2(\omega_c + \omega_m)t + \phi] + \frac{1}{2}K\cos(\omega_m t - \phi) \tag{4.16}$$

When the term at frequency $2\omega_c + \omega_m$ is filtered off, the remaining components can be written as

$$V_{out} = \frac{1}{2}K[\cos\ \omega_m t \cos\ \phi + \sin\ \omega_m t \sin\ \phi)] \tag{4.17}$$

If $\phi = 0$ (no phase error), the output is $0.5K\cos\ \omega_m t$ and when $\phi = 90^0$, the output is $0.5K\sin\ \omega_m t$. Hence in contrast to DSB-SC AM, there is also an output (shifted by 90^0 when $\phi = 90^0$. For human listeners, whose hearing is relatively insensitive to phase distortion, the requirement of phase coherence may be relaxed.

4.2.4 VSB AM Generation and Detection

VSB AM modulation is produced by filtering the corresponding DSBC AM signal. In a television transmission system, the required filtering is usually achieved by a series of cascaded high-frequency tuned amplifiers. VSB AM is virtually exclusive to television systems, and television receivers use a conventional *diode detector* to demodulate the video.

4.3 AM Comparison

1. DSBC AM systems have the advantage of simpler detectors at the receiver and thus are a natural choice where the emphasis is on inexpensive receivers. However, these require more power to transmit the same information than suppressed-carrier systems.

2. The basic difficulty with the use of DSB-SC or SSB-SC is the need to re-insert a carrier signal at the receiver, which requires the same frequency as the original suppressed carrier at the transmitter and is in phase coherence with it.

3. The primary benefit of SSB modulation is that it occupies less than half the frequency bandwidth of the DSB modulation. This helps to conserve frequency spectrum and allows more signals to be transmitted in the same frequency band.

4. Another benefit of SSB modulation is that the same information signal level can be transmitted with reduced transmitter power. This makes it possible to produce a smaller and lighter transmitter that will produce the same signal strength at the receiver as is achieved with conventional AM.

5. Because SSB signals occupy less bandwidth, reduced noise input at the receiver is also a benefit.

6. Another advantage is that SSB is not subject to selective fading. Selective fading happens when each sideband is received at a different phase, causing distortion. This cannot happen with SSB because there is only one sideband.

7. The main disadvantage of SSB modulation is that it is harder to recover the transmitted information at the receiver. The information in an AM signal is equal in frequency to the difference between the carrier and sideband frequencies. Therefore, demodulation depends on the carrier being present. If the carrier is not present, then it must be reinserted at the receiver. The reinserted carrier must have exactly the same frequency and phase as the original carrier, which is difficult to accomplish. Therefore a low-level carrier signal, called a pilot carrier, is normally transmitted along with the SSB signal. This pilot carrier is amplified at the receiver and reinserted so that the original information signal can be recovered. Because the transmitted pilot carrier has such a low power level, the benefits of SSB are still retained.

8. VSB AM modulator filter design is simplified because a sharp cut-off characteristic is not required. It also has a response which extends down to low frequencies and can include dc.

4.4 FM Generation and Detection

There are two types of frequency modulator circuits: *direct FM* and *indirect FM*. A direct FM modulator causes the frequency of a voltage-controlled oscillator (VCO) to be proportional to the modulating signal amplitude. An indirect FM modulator causes the phase of the carrier to be proportional to the amplitude of a modulating signal, resulting in frequency modulation indirectly.

In order to produce direct FM, a way must be found to vary the frequency of the carrier oscillator according to the modulating signal amplitude. A widely used integrated circuit VCO for generating a frequency-modulated carrier using direct FM is the popular 566. Also, a 555 timer IC wired as a VCO as shown in Figure 4.11 can be used to demonstrate frequency modulation using direct FM. This oscillator will generate a *square wave output*, which is acceptable as a carrier.

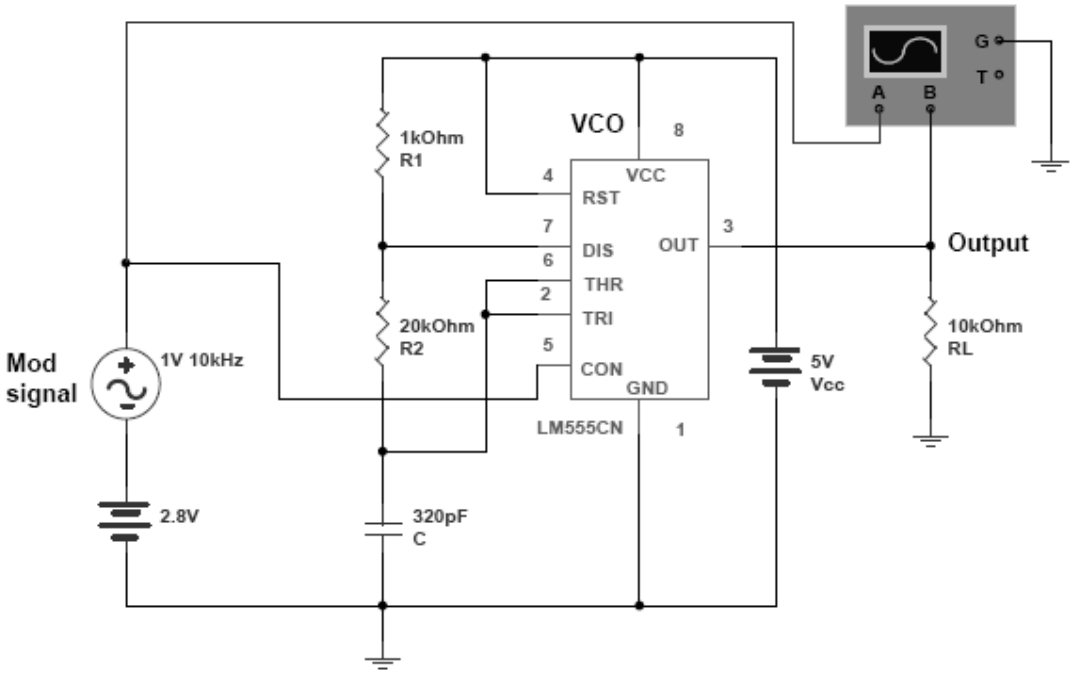


Figure 4.11: Direct FM Generation

The modulating signal voltage is applied to the control input (CON) of the 555 timer to vary the output frequency. The modulating signal voltage is riding on a 2.8 V dc voltage because the control input requires a positive voltage level at all times. *A higher control voltage on the 555 timer causes the output frequency to decrease and a lower voltage causes the output frequency to increase. Therefore, the carrier frequency will be the inverse of the modulating signal voltage level.*

The primary disadvantage of direct FM modulators is that a relatively unstable oscillator must be used to produce the variable-frequency carrier. This will cause the carrier frequency to be unstable. Such instabilities cannot be tolerated in most modern communications systems.

It has been shown that FM is equivalent to PM by the time integral of the modulating signal. Narrowband PM is itself equivalent to DSB-SC AM with carrier reinserted in phase quadrature (90^0 phase shift). The Armstrong indirect frequency modulator combines both these properties, and a block diagram is given in Figure 4.12.

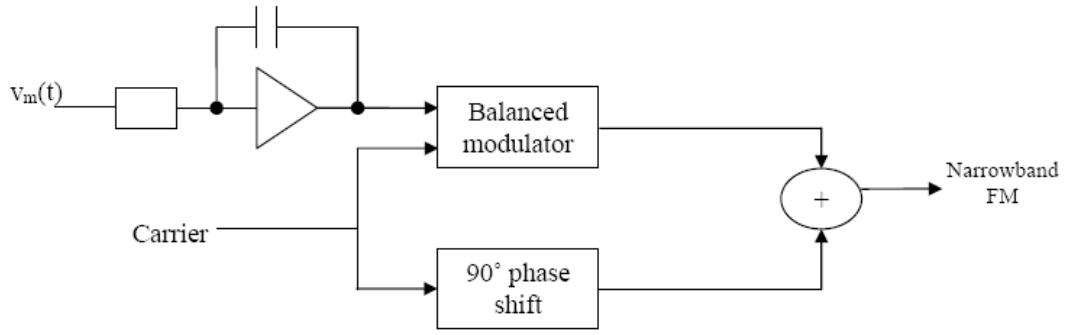


Figure 4.12: Indirect FM Generation

Any circuit that will convert the frequency variations of the carrier back to the modulating signal voltage variations can be used as an FM demodulator. FM demodulators are often called *FM detectors* or *FM discriminators*. Many different FM demodulator circuits have been developed over the years. The simplest is the slope detector, shown in Figure 4.13. The slope detector utilizes the slope of the LC resonant circuit's frequency response to convert frequency variations into voltage variations. By tuning the LC circuit resonant frequency just above the carrier frequency, the carrier frequency falls on the left side of the tuned circuit output frequency response curve. As the carrier frequency increases, the

amplitude of the tuned circuit output increases. As the carrier frequency decreases, the amplitude of the tuned circuit output decreases. Therefore, the tuned circuit output amplitude depends on the frequency of the FM carrier.

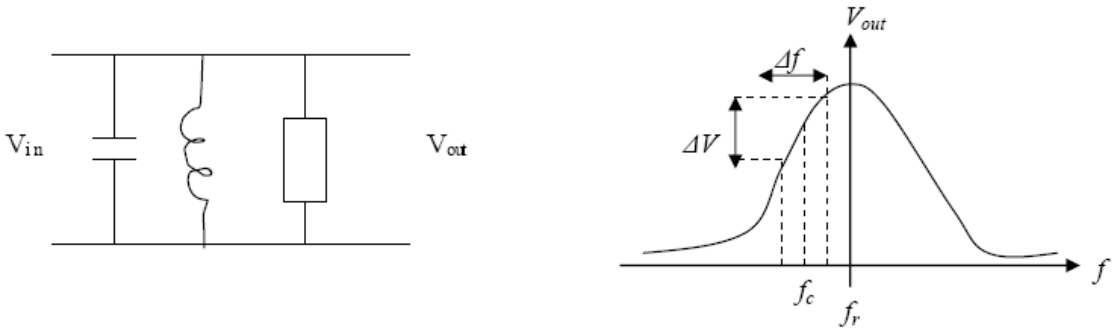


Figure 4.13: Detection of FM with a tuned circuit

The main disadvantage of the slope detector is that the tuned LC circuit is linear over only a narrow range of frequency variations, making it not very useful for wideband FM.

The phase-locked loop FM detector produces less distortion over a wider frequency deviation. For this reason, it is the most popular FM demodulator. The phase-locked loop is one of the most useful circuits in modern electronic communications because of its versatility and numerous communications applications. The PLL, shown in Figure 4.14, consists of a voltage-controlled oscillator (VCO), a phase detector, and a low-pass filter (LPF).

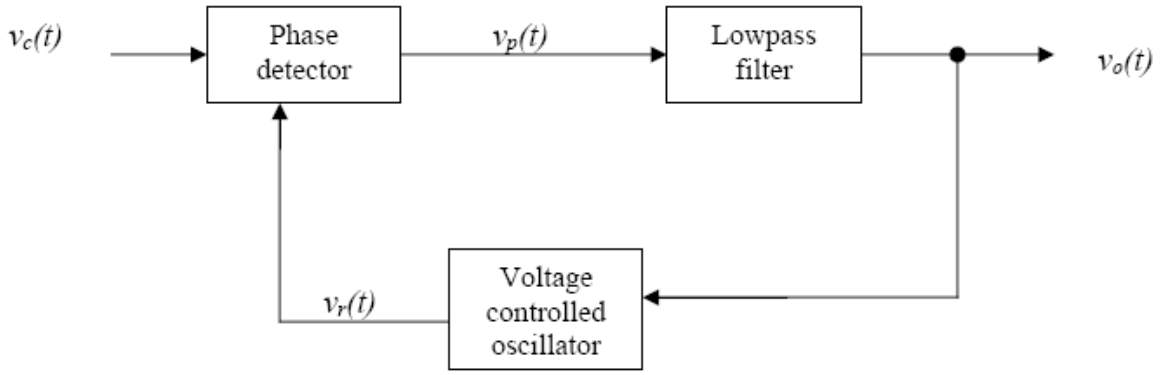


Figure 4.14: PLL as an FM Detector

It is a closed loop feedback system that locks the frequency of the VCO onto an input reference frequency. If the input frequency is modulated by a voltage and the output frequency of the VCO tracks the variation in the input frequency, then the voltage at the VCO input must be equal to the voltage which produced the frequency modulation. The phase locked loop therefore demodulates the frequency modulated input. It is the purpose of the LPF to smooth the pulsating output of the phase detector into a dc control voltage that is equal to the average output voltage of the phase detector.

Chapter 5

Multiplexing

5.1 Introduction

Multiplexing is a communications technique whereby a number of independent signals can be combined to share a common communication channel. The reverse process is called *demultiplexing*. The process can be performed either in frequency or in time. From our discussion of amplitude modulation, we have seen that it is possible to send several signals simultaneously by choosing a different carrier frequency for each signal. These carrier frequencies are chosen so that the signal spectra are not overlapping. This mode of transmission is called *frequency-division multiplexing* (FDM). The use of time for combining signals is known as *time-division multiplexing* (TDM).



Figure 5.1: Multiplexing and Demultiplexing

5.2 Frequency Division Multiplexing - FDM

Frequency division multiplexing is the positioning of signal spectra in frequency such that each signal spectrum can be separated out from all the others by filtering. Each signal in a frequency division multiplex system occupies part of transmission bandwidth for the whole of the transmission time.

Each signal modulates a sinusoidal carrier of different frequency, single sideband (SSB) modulation being used. The signals are therefore effectively stacked one above the other throughout the transmission bandwidth. To facilitate separation of the signals at the far end of the link, adjacent signals are separated by a guard band f_g (Figure 5.2).

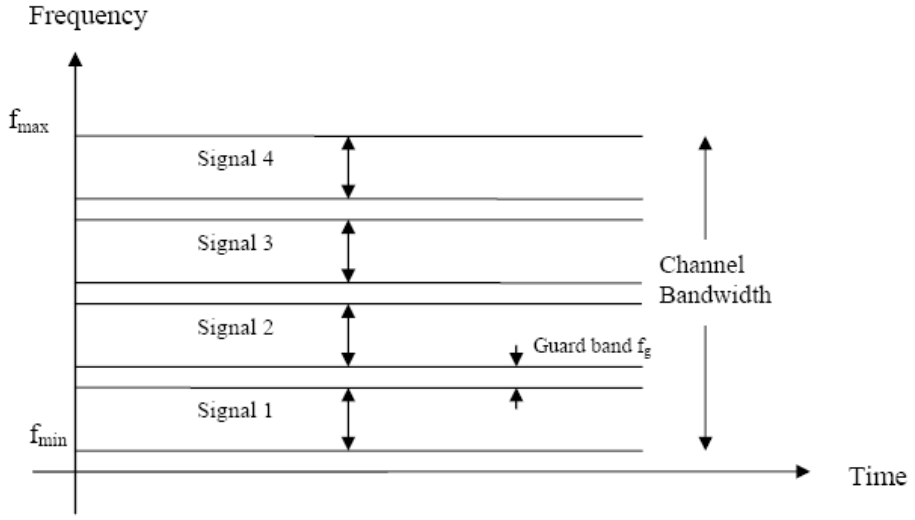


Figure 5.2: Frequency Division Multiplexing

5.2.1 FDM Systems

To illustrate the principle of FDM, let us assume that there are N signals and, for convenience, that each is bandlimited to ω_m. To separate these signals in frequency, each signal modulates a *sub-carrier* $\omega_1, \omega_2, \ldots, \omega_N$. For example, if SSB is used then each modulated signal is separated by at least ω_m, and $2\omega_m$ if DSB is used. Figure 5.3 illustrates the case in which three signals are translated in frequency and transmitted simultaneously.

Clearly the sub-carrier frequencies are selected so that each sub-carrier is separated by these minimum amounts. In practice, however, the FDM signals cannot be recovered if they overlap, and if overlap occurs, information from one channel would cross-talk into adjacent channels. To eliminate cross-talk, the sub-carrier frequencies are separated by the bandwidth required for each modulator ω_m plus a guard band frequency ω_g. Thus, the total bandwidth for an FDM system with SSB sub-carrier modulation is

$$B = Nf_m + (N - 1)f_g = N(f_m + f_g) - f_g \tag{5.1}$$

while the bandwidth for an FDM with DSB sub-carrier modulation is given by

$$B = N(2f_m) + (N - 1)f_g = N(2f_m + f_g) - f_g \tag{5.2}$$

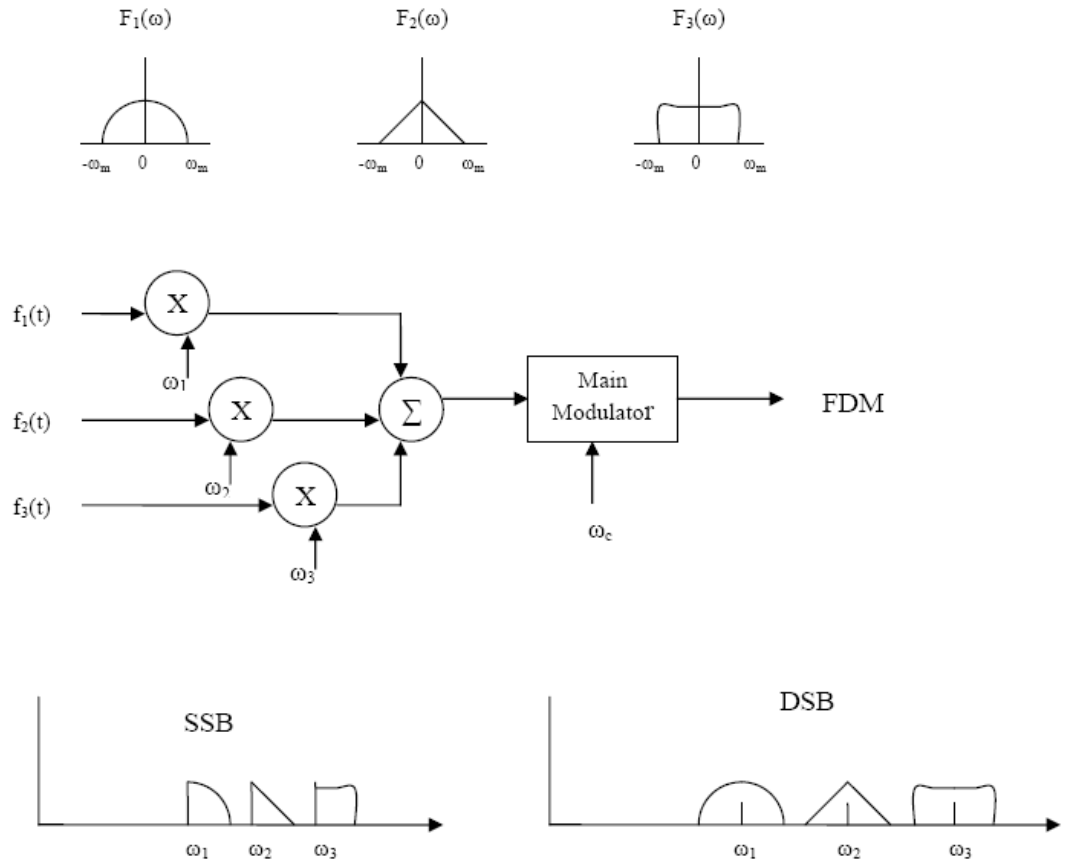


Figure 5.3: FDM Modulator

A typical FDM demodulator is shown in Figure 5.4. The main bandpass filter limits the noise from the communication channel and passes through the modulated FDM signal. This signal is demodulated to produce the FDM signal. Each bandpass filter passes its own sub-carrier components and these are demodulated by each to recover the original source signal.

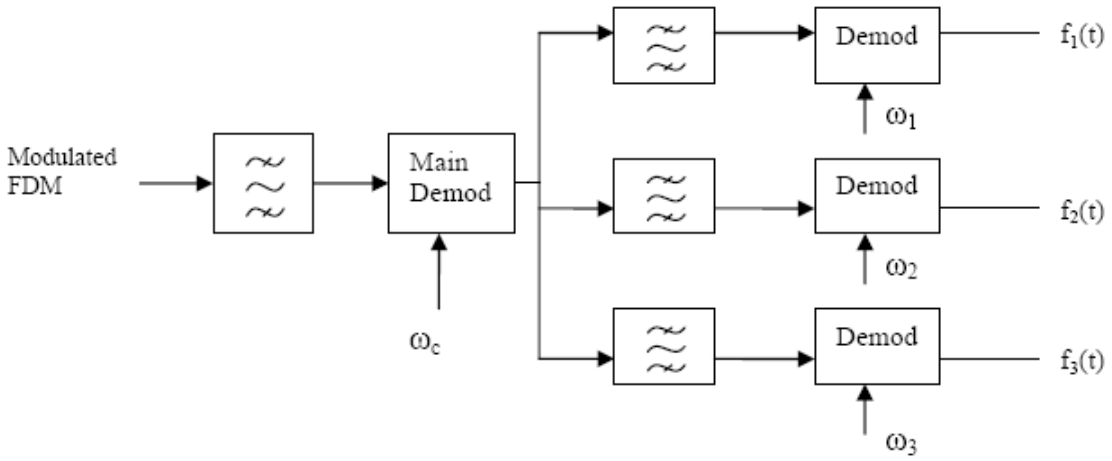


Figure 5.4: FDM Demodulator

An interesting example of the application of FDM is for multi-channel telephony.

5.2.2 Multichannel Telephony

FDM is the form of transmission used extensively in long-distance telephone systems for the simultaneous transmission of several separate telephone signals over a wideband link. Each of the telephone signals has a bandwidth limited to 4 kHz and therefore many such signals can be transmitted over, say, a coaxial cable with a bandwidth of several megahertz (4 MHz).

The complete system which can be extended up to 900 channels has a hierarchical structure shown in the table below.

Description	Number	Channels
1 channel	= 24 telegraph sub-channels	1 channel = 4 kHz
1 basic group	= 12 channels	12 channels = 48 kHz
1 super group	= 5 groups	60 channels = 240 kHz
1 master group	= 5 super groups	300 channels = 1.2 MHz
1 jumbo group	= 3 master groups	900 channels = 3.6 MHz

Figure 5.5: Basic FDM Building Blocks

CCITT (Consultative Committee for International Telephone and Telegraphy) recommends which frequencies are to be used as carriers. Multiplexing is based on a group of 12 telephone channels, called *Basic Group*. Twelve equally spaced carriers ($f_c = 112 - 4n$ kHz, where $n = 1, 2, 3, ..., 12$) are used to convert twelve analogue incoming signals to *lower-sideband only SSB-SC* signals of 4 kHz bandwidth each. The result of linearly combining the twelve bandlimited speech signal, each covering the same frequency range, is a 48 kHz wide FDM signal (60 kHz - 108 kHz) called the basic group.

Telephone channels are bandlimited and each channel is restricted within the frequency range 300 Hz - 3.4 kHz. For multiplexing purpose, a frequency bandwidth of 300 Hz - 4 kHz is allocated to provide a guard band and facilitate recovery with ease by means of filters. The twelve carrier frequencies used for the baseband signals cover the range 64 - 108 kHz with equal 4 kHz separation between each carrier (64, 68, 72, 76, 80, 84, 88, 92, 96, 100, 104 and 108 kHz). The basic group (from lower sidebands) occupies the band 60 - 108 kHz (see Figure 5.6).

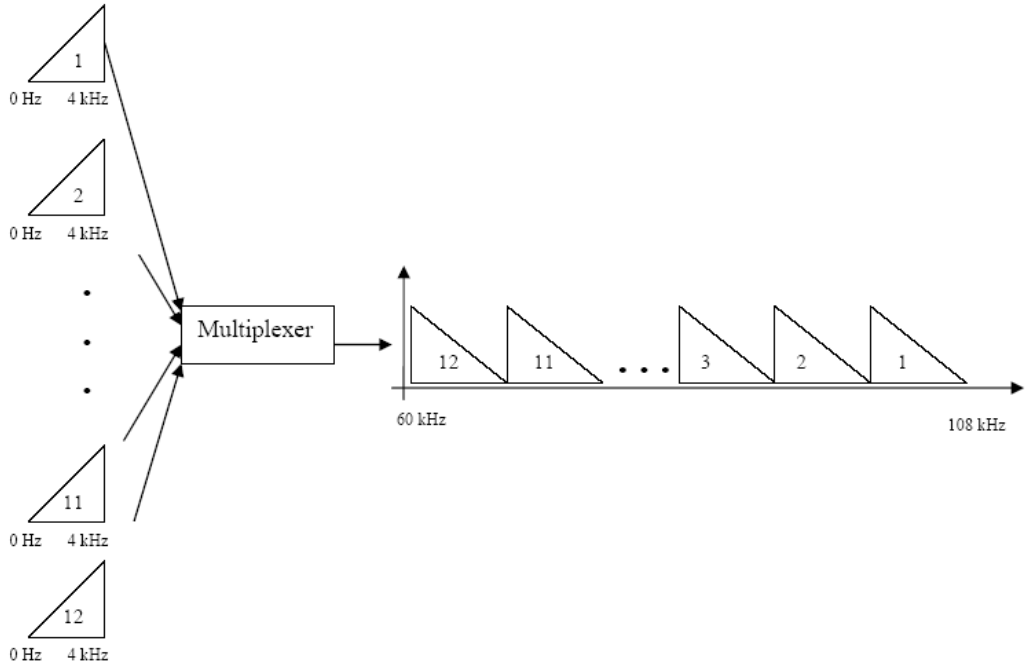


Figure 5.6: Basic group of 12 channels

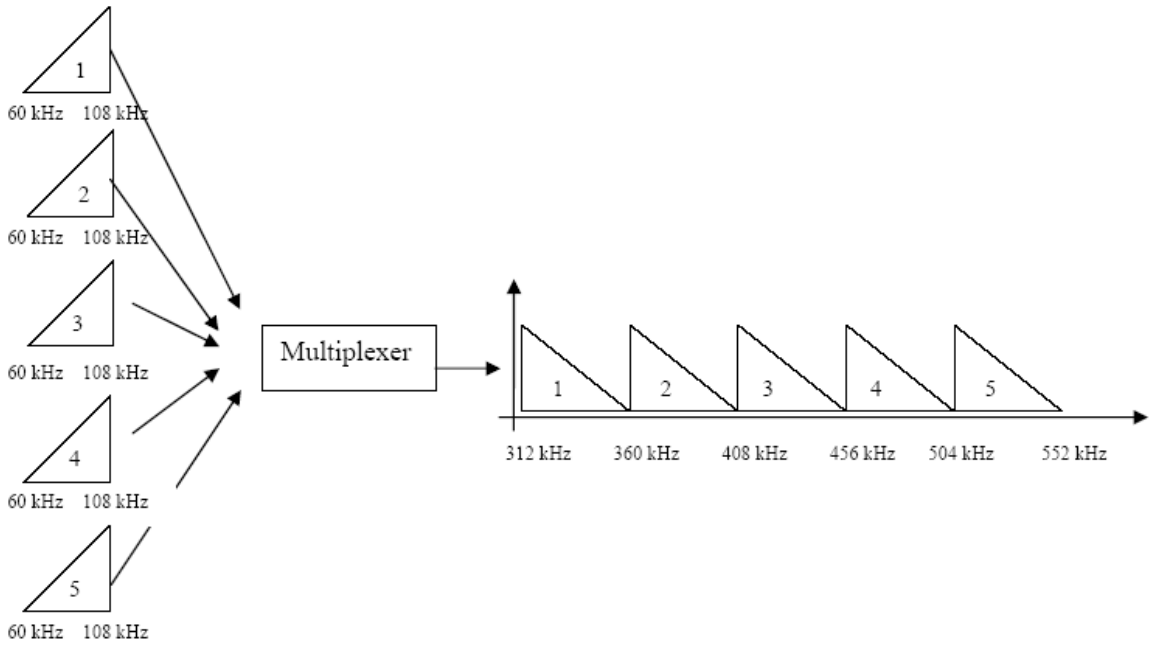


Figure 5.7: Super group from 5 basic groups

More basic groups, each of twelve channels and carrier frequencies 64 - 108 kHz, are produced in the same way as above. Five separate basic groups (i.e. 60 baseband channels) are selected for translation up in the frequency spectrum by SSB-SC (lower sideband) and linearly combined to produce a *super group* using carrier frequencies 420, 468, 516, 564 and 612 kHz (note that the carrier frequencies are separated by 48 kHz which is the bandwidth of each of the basic groups). The bandwidth of the FDM signal forming the super group is 240 kHz (312 - 552 kHz) as illustrated in Figure 5.7.

Further multiplexing to accommodate more channels (greater than 60) is done by multiplexing these super groups into a '*hyper group*' or '*master group*'. Fifteen super groups can also be multiplexed to form a '*jumbo group*' producing a 900-channel FDM system (shown in Figure 5.9).

Supergroup no.	Carrier Frequency in kHz
1	612
2	None
3	1116
4	1364
5	1612
6	1860
7	2108
8	2356
9	2604
10	2852
11	3100
12	3348
13	3596
14	3844
15	4092

Figure 5.8: Carrier frequencies: Super group to Jumbo group

Note that super group no. 2 is already in the correct frequency slot, so it does not require a modulator.



Figure 5.9: A 900-channel FDM system

5.2.3 Quadrature Multiplexing

Using the orthogonality of sines and cosines makes it possible to transmit and receive signals simultaneously on the same carrier frequency. A scheme for doing this, known as *quadrature multiplexing*, is shown in Figure 5.10. Each signal can be recovered by synchronous detection of the received signal using carriers of the same frequency but in phase quadrature.

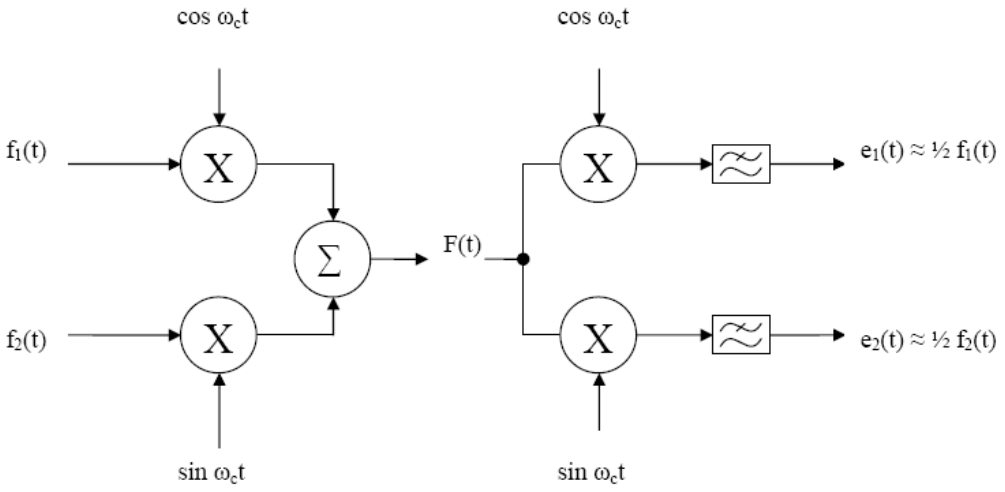


Figure 5.10: The Quadrature Multiplexing System

The transmitter output is given by

$$F(t) = f_1(t) \cos \omega_c t + f_2(t) \sin \omega_c t \tag{5.3}$$

The two receiver outputs are given by:

$$F(t) \cos \omega_c t = f_1(t) \cos^2 \omega_c t + f_2(t) \sin \omega_c t \cos \omega_c t = \frac{1}{2} f_1(t) + \frac{1}{2} f_1(t) \cos 2\omega_c t + \frac{1}{2} f_2(t) \sin 2\omega_c t \tag{5.4}$$

$$F(t) \sin \omega_c t = f_1(t) \cos \omega_c t \sin \omega_c t + f_2(t) \sin^2 \omega c t = \frac{1}{2} f_1(t) \sin 2\omega_c t + \frac{1}{2} f_2(t) - \frac{1}{2} f_2(t) \cos 2\omega_c t \tag{5.5}$$

In the lowpass filter, all terms at $2\omega_c$ are attenuated, yielding

$$e_1(t) = \frac{1}{2} f_1(t) \tag{5.6}$$

$$e_2(t) = \frac{1}{2} f_2(t) \tag{5.7}$$

Quadrature multiplexing is an efficient method of transmitting two message signals within the same bandwidth. It requires precise phase synchronization of transmitter and receiver. Quadrature multiplexing is used in transmission of colour (chroma) information signals in commercial television broadcasts.

5.2.4 FM Multiplexing

It is common practice in data transmission to combine several channels of modulated signals using frequency multiplexing methods and then modulate a high-frequency carrier with the composite multiplexed signal. To do this, the individual data signals each modulate an assigned sub-carrier. These sub-carriers are arranged so that the channels occupy adjacent frequency bands with some frequency space between them (known as guard bands). The modulated sub-carriers are used to angle-modulate a high-frequency carrier, as shown in Figure 5.11.

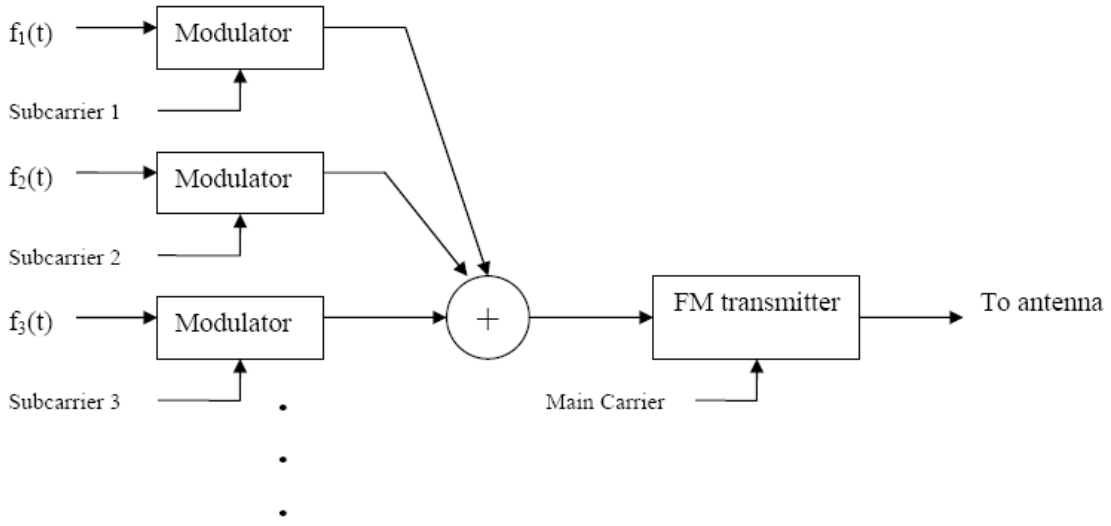


Figure 5.11: FM Multiplexing

If FM is used for the sub-carrier modulation and for the main carrier modulation, the composite modulation is referred to as *FM-FM*. If AM is used for the sub-carrier modulation and FM for the main carrier modulation, the composite modulation is referred to as *AM-FM*. The AM methods used for the sub-carrier modulation are DSB-SC or SSB-SC. Large-carrier method (DSBC) is avoided because too much of the peak frequency deviation would be used merely to send the AM carrier. Note that the stereo multiplexing used in commercial FM is an example of an AM-FM system.

5.3 Time Division Multiplexing

Time-division multiplexing (TDM) is used mainly for digital communication. In TDM, each information signal is allowed to use all the available bandwidth, but only for part of the time. Continuously varying signals, such as analog audio, are not well adapted to TDM because the signal is present at all times. On the other hand, sampled audio is very suitable for TDM, as it is possible to transmit one sample from each of several sources sequentially, then send the next sample from each source, and so on. When transmitted over a link with a fixed bandwidth, the samples spread and can overlap adjacent samples. To minimize this, *guard intervals* are allowed between adjacent signals.

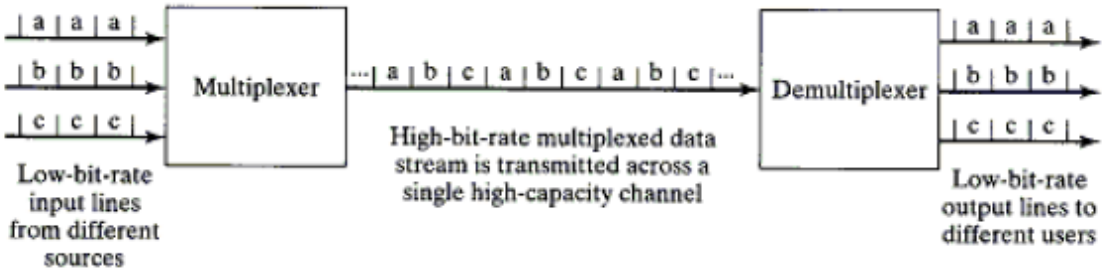


Figure 5.12: Time Division Multiplexing

Consider, for instance, a channel capable of transmitting 192 kbps. Suppose that three sources, each have 64 kbps of data that they want to transmit to individual users. As shown in Figure 5.12, the high-bit-rate channel can be divided into a series of time slots, and the time slots can be alternately used by the three sources. The three sources are thus capable of transmitting all of their data across the single, shared channel. Clearly, at the other end of the channel, the process must be reversed (i.e., the system must divide the 192 kbps multiplexed data stream back into the original three 64 kbps data streams, which are then provided to three different users).

Example: Consider the case of three streams with bit rates of 8 kbps, 16 kbps, and 24 kbps, respectively. We want to combine these streams into a single high-speed stream using TDM. The high-speed stream in this case must have a transmission rate of 48 kbps, which is the sum of the bit rates of the three sources. To determine the number of time slots to be assigned to each source in the multiplexing process. we must reduce the ratio of the rates, 8 : 16 : 24, to the lowest possible form, which in this case is 1 : 2 : 3. The sum of the reduced ratio is 6, which will then represent the minimum length of the repetitive cycle of slot assignments in the multiplexing process. The solution is now readily obtained: In each cycle of six time slots we assign one slot to Source A (8 kbps), two slots to Source B (16 kbps), and three slots to Source C (24 kbps).

So far we have considered a form of TDM that is based on fixed slot assignments to each of the low-bit-rate data streams. In other words, each stream has predefined slot positions in the combined stream, and the receiver must be aware which slots belong to which input stream. Both the transmitter and the receiver must be perfectly synchronized to the slot period. For this reason, the technique is usually called *synchronous TDM*. Note that time slots are allocated even if there is no data transmitted (idle condition) from a given source, resulting in a waste of time slots.

5.3.1 Statistical TDM

There is another important version of TDM, usually referred to as *statistical TDM*. Statistical TDM allocates time slots dynamically based on demand. For bursty data, synchronous TDM is inefficient because when any channel does not have data to transmit, "idle signal" is sent instead and no information is carried on this idle signal. This reduces the efficiency (amount of information sent per unit time) of the communication system.

For statistical TDM, instead of sending an idle signal, the system moves to the next source that has data to transmit and sends the data for that source immediately. This strategy means that capacity is not used to send idle signals. However, there is a cost. The next source with data transmits immediately; this means that the multiplexed frames can no longer be demultiplexed simply by

counting bits. We no longer have a predefined sequence of sources (slots) in each frame. Therefore, it is necessary to indicate the source of the data in each slot so that the demultiplexer can determine where to send the data from that slot.

Prior to transmission, each stream of bits coming from a source is divided into fixed-size *blocks*. Then a small group of bits called a *header* is added to each block, with the header containing the addresses of the source and intended user for that block. The block and the header are then transmitted together across the channel. Combined, the block and header are called a *packet*.